Structural Cryptanalysis of QR-UOV: Projective Direct Forgery, Equivalent-Key Recovery, and Limits of Vinegar Retuning

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Abstract

QR-UOV compresses Unbalanced Oil and Vinegar (UOV) by representing variables in a degree-three quotient ring. In this paper, we give two complementary structural attacks whose nonlinear solver cores respond oppositely to the vinegar count. For every vinegar count and every nonzero verification target, one of these attacks has solver core dimension at most $m-4$, where m is the expanded output dimension. Thus changing only the vinegar count cannot move QR-UOV beyond this fixed-output algebraic ceiling.

The direct attack sends a chosen target to $(0,\dots,0,1)$, constructs an $(m-3)$-dimensional common isotropic space for three zero-target quadrics, and quotients the remaining homogeneous system by scaling. The resulting square projective cores have dimensions 50, 74, and 101. The equivalent-key attack recovers the hidden oil graph from one regular directional root.

To estimate the cost of these cores, we split the homotopy before lifting and replace a global primitive-element requirement by local separation of one eligible simple target branch. A generic characteristic polynomial and its coefficient polars remain valid even when the chosen linear form collides on other branches. On a simple target factor, a single Frobenius sketch is exact. In the inherited Boolean-gate model, the resulting costs are 140.383995, 191.027133, and 247.125439 bits, respectively 2.616, 15.973, and 24.875 bits below the NIST category targets. Using the parent paper's weaker root-probability bound still gives 140.531461 bits at Level I.

We also implement the complete local-separator pipeline on reduced QR-UOV parameters. From only the public key, message, and salt, the program constructs the residual system, lifts and recombines all branches, performs rational specialization and exact Frobenius descent, reconstructs one point, and outputs a forgery accepted by the unchanged public verifier. Five deterministic instances and a supplementary larger-field instance all succeed and replay from public transcripts. This closes the end-to-end composability objection at toy scale, but it does not validate the full-parameter time or memory extrapolation. The Level-I statement remains conditional on the stated characteristic-polynomial degree transfer, the fast online-product schedule, the public-random expansion model, and astronomical symbolic memory.

1 Introduction

NIST opened its Additional Digital Signatures process to broaden the set of post-quantum signatures beyond the structured-lattice schemes selected in its first standardization process. Following the second round, NIST advanced nine candidates, including QR-UOV, to a third round of public analysis and permitted the teams to update their specifications and implementations [4, 5].

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QR-UOV is attractive in this setting because it retains the simple signing procedure and short signatures of Unbalanced Oil and Vinegar (UOV) while addressing UOV's main efficiency drawback: its large public key [1, 2].

UOV publishes a quadratic map from $v + m$ variables to m outputs over a finite field. A secret change of coordinates reveals two classes of variables, *vinegar variables and oil variables*, with v of the former and m of the latter. The central equations contain no oil-oil products. Once the vinegar variables are fixed, signing reduces to solving a linear system in the oil variables [3]. The same hidden structure creates two distinct attack goals. A direct forgery needs only one preimage of a verification target. An equivalent-key attack instead recovers any decomposition that can invert every target accepted by the public verifier.

The original QR-UOV construction compresses the public key by arranging its matrices into blocks represented by elements of a quotient ring $\mathbb{F}_q[X]/(f)$ [1, 2]. The recommended parameter sets use an irreducible polynomial of degree $\ell = 3$, so this quotient ring is the field $\mathbb{F}_{q^3}$. The public map is still verified over $\mathbb{F}_q$, but its block structure exposes a smaller extension-field description with V vinegar variables and m/ℓ oil variables. In the notation used throughout this paper, the expanded base-field dimension is

$$n = \ell V + m.$$

Thus changing the vinegar count V simultaneously changes the amount of underdetermination visible to a direct attack and the dimension of the hidden extension-field system visible to a structural attack (Section 3).

These two attack directions have long been studied separately. Direct attacks reduce underdetermined MQ systems before applying a polynomial-system solver [31, 32, 30]. Structural attacks recover the hidden UOV geometry through intersection or MinRank methods [10, 11, 12], while Pébereau showed that one suitable oil vector can already determine an equivalent key [15]. The QR-UOV specification and the team's latest security response cost several such attacks for fixed parameter sets [2, 29]. What these estimates do not provide is a single structural statement explaining how the direct-forgery and equivalent-key endpoints interact when V is retuned while m remains fixed. We measure that interaction by the number of variables in the square nonlinear system passed to the root solver, which we call the *solver core*.

This brings us to the question that governs the paper:

> **Question.** *Can QR-UOV raise the nonlinear solver cores of both attacks above $m - 4$ by changing only the vinegar count while keeping the output dimension and quotient degree fixed?*

1.1 Our results and technical overview

We answer this question negatively. Our main result is the following fixed-output algebraic ceiling:

For every vinegar count and every nonzero verification target, one of two complementary attacks on QR-UOV has solver core dimension at most $m - 4$.

The statement, formalized in Theorem 4.9, concerns the algebraic bottleneck shared by the two attacks. It does not assert that their running time or memory usage depends only on this dimension.

Projective direct forgery. For a fixed nonzero target, a public output transformation makes three target coordinates zero. We use Chevalley–Warning to construct an $(m - 3)$-dimensional subspace on which the corresponding three quadratic forms vanish. The remaining zero-target equations restrict to homogeneous quadrics, and quotienting their common scaling turns the residual problem into a square projective system of dimension $m-4$. For the three recommended parameter sets, this gives solver cores of dimensions 50, 74, and 101 (Theorems 4.8 and 4.10). The construction uses public attack coins, and every returned vector is checked against the unchanged public verification map. A solver failure can therefore force a retry, but it cannot produce an accepted false signature (Section 4).

Directional equivalent-key recovery. In extension-field coordinates, the hidden oil space is the graph of a linear map G. Fixing a public direction c produces a system of m quadrics in V variables with the planted root Gc. If the derivative at a recovered root has full rank, public graph identities determine all of G. Exact pull-back and signing checks then certify an equivalent key that signs every target accepted by the verifier (Theorem 7.8). Pébereau established the equivalent-key endpoint from one suitable oil vector [15]; our contribution is a public polynomial-system mechanism for finding and certifying the required information in QR-UOV.

The fixed-output security scissors. The direct and directional routes respond oppositely to V. When $V \leq m-\ell-1$, directional recovery already has a core of dimension at most $m-\ell-1$. When $V \geq m-\ell$, target-orthogonal slicing with ℓ equations followed by projectivization gives a direct core of that same dimension. Since these cases cover every integer V, the combined algebraic core satisfies

$$r_{\text{attack}}(V) \leq m-\ell-1.$$

For QR-UOV, $\ell = 3$, so the ceiling is $m-4$ (Theorem 4.9). Increasing only V therefore cannot move the two attacks past their crossing point. A parameter revision must also increase m, or change the construction so that one of the reductions no longer applies.

Concrete estimates and executable evidence. To cost these cores, we adapt a nonsingular-root homotopy to characteristic 127, batch the dense quadratic lifting operations, choose sparse working fields, and replace global branch separation by a local separator for one eligible target branch. In the inherited Boolean-gate model, the resulting diagnostic costs are $\mathbf{2^{140.384}}$, $2^{191.027}$, and $2^{247.125}$ gates at Levels I, III, and V, respectively. These rows lie 2.616, 15.973, and 24.875 bits below their category targets (Table 6). The Level-I row is a candidate model-based break, not a practical computation: it depends on the stated characteristic-polynomial degree transfer and fast online-product schedule, and the direct symbolic representation has prohibitive memory requirements (Sections 8 and 10).

The reduced-parameter implementation executes the complete local-separator pipeline from public inputs and returns signatures accepted by the unchanged verifier over two toy fields. This experiment checks that the algebraic components compose, but it does not benchmark or validate the full-parameter symbolic schedule (Section 9). The direct and directional reductions, the fixed-output scissors, and verification of every returned signature or key are algebraic statements. The root-density arguments use the stated public-random or ideal-primitive models, while the finite gate rows remain conditional symbolic estimates. We keep these claim boundaries explicit throughout the paper.

Organization. Section 2 positions the two attacks against prior work. Section 3 connects the extension-field description to the official verification map. Section 4 proves the projective direct forgery and the fixed-output scissors, while Sections 5 to 7 develop and verify directional recovery. Section 8 gives the concrete solver refinements, and Sections 9 and 10 separate the evidence, assumptions, cost, memory, and parameter consequences.

2 Related work

Direct attacks on underdetermined MQ. Thomae–Wolf, Furue et al., and Hashimoto reduce underdetermined quadratic systems before applying an MQ solver [31, 32, 30]. Hashimoto's line is the benchmark used in the QR-UOV specification. The QR-UOV team's January 2026 response reports classical Hashimoto estimates of 158, 217, and 285 bits for the recommended sets [29]. The affine residual dimensions $51, 75, 102$ are therefore prior art. Our contribution is a self-contained homogeneous construction, a fixed-key seed-disjointness bound, a QR-UOV-specific nonsingular-root theorem, and the observation that full target normalization lets the solver quotient by radial scaling, reducing the square geometric cores to $50, 74, 101$. We then cost those projective cores with a separate finite-characteristic solver. The recent "Just Guess" algorithm improves the quantum analysis of underdetermined MQ but does not improve the team's reported classical QR-UOV estimates [14, 29].

Equivalent-key recovery for UOV. Pébereau showed that one suitable oil vector can suffice for polynomial-time UOV equivalent-key recovery [15]. We do not claim that endpoint as new. The QR-UOV-specific contribution is the public mechanism for finding the needed information: a direction exposes a planted root Gc, full derivative rank makes it isolated, and cross-direction graph identities recover all of G. Exact pull-back and signing checks then certify a key-only universal forgery.

Beullens' intersection attack and later work recover hidden UOV geometry through structured linear algebra [10]. Rectangular-MinRank estimates were sharpened for QR-UOV and related schemes [11, 12]. Truncated-polynomial-ring methods improve some UOV parameters but do not report an improvement for the QR-UOV sets studied here [13]. Odd-characteristic symmetric-algebra attacks apply to QR-UOV but, at characteristic 127, do not improve the existing QR-UOV estimates [35].

Polynomial-system solving. The arbitrary-core engine is the nonsingular-solutions homotopy of Safey El Din and Schost [27], building on symbolic homotopy and geometric-resolution methods [28, 8]. We replace the start-system and separator choices responsible for its published characteristic restriction. The appendices retain independent checks using the near-quadratic solver of van der Hoeven–Lecerf and a locally closed Kronecker algorithm [7, 9].

Chevalley–Warning. The direct reduction uses the classical fact that, when the sum of the degrees is smaller than the number of variables, the number of common zeros over a finite field is divisible by the characteristic [33, 34]. Applied to three quadrics in seven variables, it guarantees a nonzero extension vector. Exhaustive projective search makes the theorem constructive, so no unpriced existence oracle is hidden in the preprocessing.

3 Expanded-key model and exact signing certificates

This section connects the algebra used by the attack to the official verification map. There are two coordinate systems:

- the official base-field system over $\mathbb{F}_{127}$, in which signatures are verified; and
- a smaller extension-field system over $K = \mathbb{F}_{127^3}$, in which the hidden oil space and the attack are easier to describe.

The specification gives an exact, invertible pull-back between them. We first recover and verify the hidden structure over K, then pull it back and check the resulting base-field factorization against the unchanged public key. One invertible oil system then certifies that the recovered decomposition can sign every target.

The Round-2 $q = 127$ parameter sets use quotient-ring degree $\ell = 3$. The specification has $v = \ell V$ base-field vinegar variables and $m = \ell O$ base-field oil variables and output coordinates. After quotient-ring expansion, the attacker works with the same m outputs as quadratic forms in only $N = V + O$ variables over

$$K = \mathbb{F}_{q^\ell} = \mathbb{F}_{127^3}, \qquad Q = |K| = 127^3 = 2{,}048{,}383, \qquad \log_2 Q = 20.966054\ldots.$$

Thus m denotes both the number of public output coordinates and the base-field oil dimension, whereas $O = m/\ell$ is the oil dimension over K.

Let $\Phi_f : K \to \mathbb{F}_q^{\ell\times\ell}$ be the regular representation from the specification, and let $W_f \in \mathbb{F}_q^{\ell\times\ell}$ be its invertible symmetrizer. Write ϕ for the blockwise inverse of Φ_f, and write $W_f^{(a)}$ for the block-diagonal matrix with a copies of W_f.

The attack uses the expanded extension-field tuple

$$P = (p_1, \ldots, p_m), \qquad p_i(x) = x^\mathsf{T} P_i x, \qquad x \in K^N,$$

where the symmetric matrices P_i share a hidden totally isotropic O-space $\mathcal{O}$:

$$u^\mathsf{T} P_i w = 0 \qquad (u, w \in \mathcal{O},\ 1 \le i \le m). \tag{1}$$

The Round-2 sets analyzed in this paper are

Level	q	ℓ	v	m	$V = v/\ell$	$O = m/\ell$
I	127	3	156	54	52	18
III	127	3	228	78	76	26
V	127	3	306	105	102	35

Symbol	Meaning
$N = V + O$	Number of extension-field variables.
$e = m - V$	Number of public equations not used in the square V-equation core.
$D = 2^V$	Bézout degree of a smooth projective core.
$L_s = \mathbb{F}_{Q^s}$	Solver working field; the same field supplies slicing, recombination, and rational-root extraction.

All concrete theorems below concern these three sets, for which $m < q$. The directional and projective equivalent-key analyses use Theorem 3.1 and allow arbitrary correlation between the secret graph and the symmetric public blocks. The direct-forgery root-counting theorem uses the stronger independent-graph specialization of Theorem 3.2. The fastest projective route has cost $\widetilde{O}(2^V 2^{(1+\varepsilon)(V-1)} 3 \log_2 127)$ for every fixed rational $\varepsilon > 0$, while the graph-direction route uses the explicit locally closed finite-field theorem of Giménez et al. Section C gives separate ideal-XOF and ideal-cipher transfers for the two models and isolates the remaining fixed-primitive gap.

3.1 Public-random expansion model and relation to seeded expansion

Definition 3.1 (Public-random expansion with an arbitrarily correlated secret graph)**.** First sample

$$(A_1, \ldots, A_m) \leftarrow \mathrm{Sym}_V(K)^m$$

from the product-uniform distribution. Conditional on the resulting tuple $A = (A_i)_i$, sample the secret graph $G \in K^{V\times O}$ according to an arbitrary conditional distribution. Conditional on (A, G), sample

$$(E_1, \dots, E_m) \leftarrow (K^{V\times O})^m$$

from the product-uniform distribution, set each lower-right public block by the key-generation identity, and expose the resulting expanded public tuple. Values are sampled once and reused consistently. No independence between G and A is assumed.

Definition 3.2 (Independent-graph specialization)**.** The *independent-graph public-random model* is Theorem 3.1 with the additional requirement that the marginal law of G be independent of A. Equivalently, after centralization the product-uniform coefficient pairs $(A_i, B_i)_{1\le i\le m}$ are independent of G. The marginal law of G may otherwise be arbitrary.

Proposition 3.3 (Exact central-block factorization under graph correlation)**.** In the block notation of (2)–(4), define

$$B_i := E_i - A_i G.$$

Conditional on every fixed pair (A, G), the tuple $(B_1, \dots, B_m)$ is product-uniform and its conditional law does not depend on (A, G). Consequently, B is independent of (A, G), and the pairs $(A_i, B_i)_{1\le i\le m}$ are independent and uniform on $\mathrm{Sym}_V(K) \times K^{V\times O}$. The graph G may nevertheless remain arbitrarily correlated with A. The lower-right public blocks C_i are determined by (4).

Proof. Fix $(A, G) = (a, g)$. For each i, the map

$$E_i \longmapsto B_i = E_i - a_i g$$

is a translation of the additive space $K^{V\times O}$ and hence a bijection. It sends the product-uniform law of E to the product-uniform law of B, independently of (a, g). Thus B is independent of (A, G). Since A itself is product-uniform, $(A_i, B_i)_i$ has the claimed product law. The formula for C_i is the key-generation block identity (4). □

The directional and projective bad-key calculations depend on the central pairs (A_i, B_i), uniform public slices, and fresh attack coins. A secret coordinate change depending on G maps a uniform public slice to a uniform central slice for every fixed key, so those calculations tolerate arbitrary G–A correlation. The direct-forgery reduction itself is also key by key, but its average-case residual-root theorem conditions on an off-oil point and then uses independence of G from the central coefficient pairs; it is therefore stated under Theorem 3.2. Section C proves separate ideal-primitive transfers, including the public/secret seed-collision term needed for the independent-graph model. All returned keys and signatures are verified against the unchanged public map, independently of any distributional model.

3.2 Central and public graph coordinates

In secret central coordinates $K^V \oplus K^O$,

$$F_i = \begin{pmatrix} A_i & B_i \\ B_i^\top & 0 \end{pmatrix}, \qquad F_i(v, o) = v^\top A_i v + 2v^\top B_i o, \tag{2}$$

with $A_i \in \mathrm{Sym}_V(K)$ and $B_i \in K^{V\times O}$ independent and uniform. The planted oil space is $\mathcal{O}_0 = \{0\} \oplus K^O$.

In normalized public coordinates the hidden space is a graph

$$\mathcal{O} = \left\{ \begin{pmatrix} -Gc \\ c \end{pmatrix} : c \in K^O \right\}. \tag{3}$$

Writing
$$P_i = \begin{pmatrix} A_i & E_i \\ E_i^{\mathsf{T}} & C_i \end{pmatrix},$$
total isotropy is equivalent to
$$C_i = G^{\mathsf{T}} E_i + E_i^{\mathsf{T}} G - G^{\mathsf{T}} A_i G. \tag{4}$$

Proposition 3.4 (Verified graph gives an exact expanded UOV decomposition)**.** Let $G' \in K^{V\times O}$ satisfy every identity (4), and put
$$B_i' = E_i - A_i G', \qquad F_i' = \begin{pmatrix} A_i & B_i' \\ B_i'^{\mathsf{T}} & 0 \end{pmatrix}, \qquad S_{G'} = \begin{pmatrix} I_V & G' \\ 0 & I_O \end{pmatrix}.$$
Then
$$P_i = S_{G'}^{\mathsf{T}} F_i' S_{G'} \qquad (1 \le i \le m).$$
Consequently, the specification's inverse pull-back gives an expanded base-field UOV decomposition of the unchanged verification map.

Proof. Since
$$S_{G'}^{-1} = \begin{pmatrix} I_V & -G' \\ 0 & I_O \end{pmatrix},$$
a direct block multiplication gives
$$S_{G'}^{-\mathsf{T}} P_i S_{G'}^{-1} = \begin{pmatrix} A_i & E_i - A_i G' \\ E_i^{\mathsf{T}} - G'^{\mathsf{T}} A_i & C_i - G'^{\mathsf{T}} E_i - E_i^{\mathsf{T}} G' + G'^{\mathsf{T}} A_i G' \end{pmatrix}.$$
The lower-right block is zero by (4); the remaining blocks are exactly F_i'. Rearranging proves the displayed factorization.

To make the interface with the specification explicit, let hats denote the corresponding base-field matrices. The forward pull-back of Section 4.1 of the specification has the form
$$P_i = \phi\Big((W_f^{(N)})^{-1} \widehat{P}_i\Big), \qquad F_i' = \phi\Big((W_f^{(N)})^{-1} \widehat{F}_i'\Big), \qquad S_{G'} = \phi(\widehat{S}_{G'}).$$
Hence the inverse pull-back sets
$$\widehat{P}_i = W_f^{(N)} \phi^{-1}(P_i), \qquad \widehat{F}_i' = W_f^{(N)} \phi^{-1}(F_i'), \qquad \widehat{S}_{G'} = \phi^{-1}(S_{G'}).$$
The pull-back identity in the specification converts the verified extension-field factorization into $\widehat{P}_i = \widehat{S}_{G'}^{\mathsf{T}} \widehat{F}_i' \widehat{S}_{G'}$ for the unchanged base-field verification map [2, Sec. 4.1]. These matrix identities publicly certify an exact expanded UOV decomposition. Signing uses this base-field pull-back, not the extension-field tuple with only $O = m/3$ oil variables. The separate test in Theorem 3.5 certifies that the decomposition can invert every target. ☐

For an expanded base-field decomposition $\widehat{P}_i = \widehat{S}^{\mathsf{T}} \widehat{F}_i \widehat{S}$, write the central variables as $(a, o) \in \mathbb{F}_q^v \oplus \mathbb{F}_q^m$. Because the oil–oil block of every $\widehat{F}_i$ is zero, fixing a leaves a linear map in the oil variables. Denote its matrix by
$$\mathsf{L}_{\widehat{F}}(a) \in \mathbb{F}_q^{m\times m}, \qquad \big(\mathsf{L}_{\widehat{F}}(a) o\big)_i := \widehat{F}_i(a, o) - \widehat{F}_i(a, 0). \tag{5}$$

Proposition 3.5 (Public signing witness)**.** If $\det \mathsf{L}_{\widehat{F}}(a) \ne 0$, then $(\widehat{F}, \widehat{S}, a)$ is a publicly checkable signing certificate. For every target $y \in \mathbb{F}_q^m$, one computes
$$o = \mathsf{L}_{\widehat{F}}(a)^{-1}\big(y - \widehat{F}(a, 0)\big), \qquad x = \widehat{S}^{-1}(a, o),$$
and obtains $\widehat{P}(x) = y$.

Proof. Direct substitution gives $\widehat{F}(a,o) = y$, followed by $\widehat{P}(x) = \widehat{F}(\widehat{S}x) = y$. These signatures verify under the unchanged public map; matching the official signing distribution is neither needed nor claimed. $\square$

Corollary 3.6 (Key-only universal chosen-message forgery)**.** Given a publicly verified signing certificate from Theorem 3.5, an attacker can forge an accepted QR-UOV signature on every chosen message without querying a signing oracle. Concretely, for a message M, choose any allowed salt r, compute the official target

$$\mu = \mathrm{SHAKE256}(\mathsf{seed}_{\mathrm{pk}} \parallel \mathrm{BytesToBits}(M), 512), \qquad y = \mathrm{Hash}(\mu, r),$$

invert y using Theorem 3.5, and output (r, x). The official verifier accepts because $\widehat{P}(x) = \mathrm{Hash}(\mu, r)$.

Proof. The specification's verifier recomputes μ and $\mathrm{Hash}(\mu, r)$ and accepts exactly when the public map evaluated at the signature vector equals that target [2, Secs. 4.2–4.3]. Theorem 3.5 inverts every target under the unchanged public map. No distributional claim about the forged signature is needed for verification, and the secret seed is never recovered. $\square$

Fix a nonzero $\xi \in \mathbb{F}_q^{\ell}$. For each quotient-ring vinegar block $1 \le j \le V$, let a_j place ξ in block j and zero elsewhere, and set

$$M_j := \mathsf{L}_{\widehat{F}}(a_j) \in \mathbb{F}_q^{m \times m}.$$

For $0 \le r \le m$, write

$$\varrho_r := \Pr_{M \leftarrow \mathbb{F}_q^{m\times m}}[\operatorname{rank} M = r] = q^{-m^2} \prod_{i=0}^{r-1} \frac{(q^m - q^i)^2}{q^r - q^i}. \tag{6}$$

In particular,

$$\pi_m := \varrho_m = \prod_{k=1}^{m} (1 - q^{-k}), \qquad \sigma_m := 1 - \pi_m. \tag{7}$$

For the stronger fallback, define

$$\chi_m := q^{-(m-1)} + (1 - q^{-(m-1)})q^{-1}, \qquad \zeta_{\mathrm{pair}} := \varrho_{m-1} q^{-1} \chi_m + (1 - \varrho_m - \varrho_{m-1})(1 - \varrho_m). \tag{8}$$

Proposition 3.7 (Basis-first deterministic signing search)**.** For the planted decomposition in the public-random expansion model, the matrices $M_1, \dots, M_V$ are independent and uniform in $\mathbb{F}_q^{m\times m}$. Consequently:

1. Testing the basis assignments $a_1, a_2, \dots$ in order uses fewer than

$$\pi_m^{-1} < 1.008$$

basis tests on average, stopping at the first success or after all V tests. It fails to find an invertible matrix exactly when all V basis matrices are singular, an event of probability

$$\eta_{\mathrm{basis}} := \sigma_m^V, \tag{9}$$

whose base-two logarithms at Levels I, III, and V are

$$-362.823, \qquad -530.280, \qquad -711.692.$$

2. Much shorter fixed lists already suffice at the category scale. Testing the first

$$b_{\text{sign}} = 22, \qquad 31, \qquad 40$$

basis assignments at Levels I, III, and V fails with probabilities $\sigma_m^{b_{\text{sign}}}$, whose base-two logarithms are

$$-153.502, \qquad -216.298, \qquad -279.095. \tag{10}$$

3. Pair the basis matrices as $(M_1, M_2), (M_3, M_4), \ldots$. Let $\mathcal{E}_{\text{sign}}$ be the event that every designated pair span contains only singular matrices. Then

$$\Pr[\mathcal{E}_{\text{sign}}] \le \eta_{\text{sign}} := \zeta_{\text{pair}}^{\lfloor V/2 \rfloor}, \tag{11}$$

with base-two logarithms below

$$-544.820, \qquad -796.276, \qquad -1068.687.$$

Outside $\mathcal{E}_{\text{sign}}$, a deterministic fallback succeeds after at most

$$V + \lfloor V/2 \rfloor (q-1) \tag{12}$$

oil-matrix tests: first test every M_j; if necessary, for each pair (A, B) test $A + tB$ for all $t \in \mathbb{F}_q^\times$. The basis tests already cover the two remaining projective points A and B.

For any fixed exact candidate, define its oil-matrix determinant polynomial

$$\Delta(c_1, \ldots, c_V) := \det\left(\sum_{j=1}^{V} c_j M_j\right).$$

Whenever Δ is nonzero, a uniform scalar-block vinegar assignment is invertible with probability at least $1 - m/q$. The corresponding expected numbers of assignments are at most

$$\frac{q}{q-m} = 1.740, \qquad 2.592, \qquad 5.773$$

at Levels I, III, and V. This last statement is key by key. For a recovered candidate not known to be planted, every successful determinant test is still an exact public signing certificate.

Proof. Write $b_{i,j,k} \in K = \mathbb{F}_{q^\ell}$ for entry (j,k) of the planted extension-field block B_i. Under the public-random experiment these elements are independent and uniform. The (i,k) block of the oil coefficient row produced by a_j is

$$2\xi^\mathsf{T} W_f \Phi_f(b_{i,j,k}) \in \mathbb{F}_q^{1\times\ell}.$$

The factor 2 is invertible. The $\mathbb{F}_q$-linear map

$$K \longrightarrow \mathbb{F}_q^{1\times\ell}, \qquad b \longmapsto \xi^\mathsf{T} W_f \Phi_f(b)$$

is injective: if $b \neq 0$, then $\Phi_f(b)$ is invertible, while $\xi^\mathsf{T} W_f \neq 0$. Equal dimensions make it an isomorphism. Thus each M_j is uniform, and distinct j use disjoint entries of the independent blocks B_i; hence the M_j are independent.

A basis test succeeds with probability π_m. Truncating a geometric random variable at V gives expected test count

$$\sum_{j=0}^{V-1} \sigma_m^j = \frac{1-\sigma_m^V}{\pi_m} < \pi_m^{-1}.$$

Independence gives σ_m^b for every fixed prefix of length b, yielding all displayed basis-list probabilities.

It remains to bound one pair. Let A, B be independent uniform $m \times m$ matrices, and let $E_{A,B}$ be the event that their span contains no invertible matrix. If A is invertible, this event is impossible. If $\operatorname{rank} A = m-1$, invertible left and right changes of basis reduce A to $\operatorname{diag}(I_{m-1}, 0)$ without changing the distribution of B. Write

$$B = \begin{pmatrix} C & b \\ c^\mathsf{T} & d \end{pmatrix}.$$

On $E_{A,B}$, the polynomial $\det(A+tB)$ vanishes for every $t \in \mathbb{F}_q$. Its degree is at most $m < q$, so it is the zero polynomial. Its linear coefficient is d, and, after $d = 0$, its quadratic coefficient is $-c^\mathsf{T} b$. These conditions have probability at most $q^{-1}\chi_m$. If $\operatorname{rank} A \le m-2$, then $E_{A,B}$ implies that B is singular, an event of probability $1 - \varrho_m$. Conditioning on the rank of A gives $\Pr[E_{A,B}] \le \zeta_{\text{pair}}$. The designated pairs are independent, proving (11).

The basis matrices and $A + tB$ for $t \in \mathbb{F}_q^\times$ enumerate every projective point on each designated pair line, so the deterministic scan succeeds whenever one pair span contains an invertible matrix. Finally, Δ has total degree m. If it is nonzero, Schwartz–Zippel gives success probability at least $1 - m/q$ for a uniform coefficient vector. □

4 Projective target-orthogonal direct forgery

The directional attack uses QR-UOV's extension-field graph. This section gives a second route that works directly with the official base-field verification map

$$\widehat{P} : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q^m, \qquad n = v + m, \qquad q = 127.$$

It does not recover a key. Given a nonzero target y, it constructs a public subspace on which three target equations disappear. Full output normalization then leaves $m-4$ homogeneous zero-target equations in projective dimension $m-4$; one final equation only recovers the affine scale. The square solver cores are

$$50, \qquad 74, \qquad 101$$

for Levels I, III, and V.

The preceding affine dimensions $51, 75, 102$ coincide with the $\alpha = 3$ Hashimoto line already used in the QR-UOV specification's direct-attack analysis [30, 2]. We do not claim those dimensions as new. The further one-variable reduction follows from two facts specific to the homogeneous verification problem: every target but one can be made zero, and the remaining zero equations are invariant under nonzero scalar multiplication. The other new ingredients are a fixed-key seed bound, a finite Chevalley–Warning construction, a second-moment lower bound for an eligible nonsingular projective root, and a characteristic-127 solver ledger independent of WXL semi-regularity estimates.

The algebraic reduction below is key by key. Its probability calculation uses the independent-graph public-random model of Theorem 3.2: the hidden graph G is independent of the product-uniform central coefficient pairs (A_i, B_i). This independence is needed only for the direct attack's average-case root analysis; the directional results continue to allow arbitrary G–A correlation. In the ideal-XOF and ideal-cipher models, official public and secret expansion have this independence whenever their independently sampled seeds differ. Section C adds the seed-collision and buffer terms explicitly. No such transfer is claimed for the fixed SHAKE or AES functions.

4.1 The pull-back still supplies full first-order randomness

The official degree-three pull-back produces a restricted family of base-field quadrics, not the set of all UOV quadrics over $\mathbb{F}_q$. We therefore record the exact local property needed below.

Let $E = K^{V+O}$, viewed as an $n = 3(V + O)$ dimensional vector space over $\mathbb{F}_q$, and let $\widehat{\mathcal{O}} \subset E$ be the hidden $m = 3O$ dimensional base-field oil space. In central coordinates, one public-random expanded form is obtained from

$$p(v, o) = v^{\mathsf{T}} Av + 2v^{\mathsf{T}} Bo$$

with $A \in \mathrm{Sym}_V(K)$ and $B \in K^{V\times O}$ uniform. The specification's pull-back evaluates a fixed nonzero $\mathbb{F}_q$-linear functional of $p(v, o)$. Equivalently, after choosing a nonzero $\gamma \in K$, it has the form

$$Q(v, o) = \mathrm{Tr}_{K/\mathbb{F}_q}(\gamma p(v, o)) \,. \tag{13}$$

The precise value of γ depends on the chosen regular-representation basis and is irrelevant; nondegeneracy of the trace pairing is what matters.

Lemma 4.1 (Evaluation and one-jet surjectivity)**.** Let $x \in E \setminus \widehat{\mathcal{O}}$, and let $U \subseteq E$ be an $\mathbb{F}_q$-linear subspace containing x. For a uniform pull-back form Q as in (13):

1. $Q(x)$ is uniform in $\mathbb{F}_q$;
2. the map
$$Q \longmapsto (Q(x), dQ_x|_U)$$
is onto
$$\{(a, \lambda) \in \mathbb{F}_q \times U^* : \lambda(x) = 2a\};$$
consequently, conditioned on $Q(x) = a$, the derivative $dQ_x|_U$ is uniform in the displayed affine hyperplane;
3. for $x, z \in E \setminus \widehat{\mathcal{O}}$, the two evaluation functionals $Q \mapsto Q(x)$ and $Q \mapsto Q(z)$ are proportional over $\mathbb{F}_q$ only if $z = \lambda x$ for some $\lambda \in K$ with $\lambda^2 \in \mathbb{F}_q$. Since $[K : \mathbb{F}_q] = 3$ is odd, this forces $\lambda \in \mathbb{F}_q$.

Proof. Write $x = (v_x, o_x)$ in central K-coordinates. Since x is off oil, $v_x \neq 0$. Choose a K-linear functional $\varphi : E \to K$ with $\varphi|_{\widehat{\mathcal{O}}} = 0$ and $\varphi(x) = 1$. The map

$$h \longmapsto (u \mapsto \mathrm{Tr}_{K/\mathbb{F}_q}(\gamma h(u)))$$

from K-linear maps $E \to K$ to $\mathbb{F}_q$-linear maps $E \to \mathbb{F}_q$ is an isomorphism, by nondegeneracy of the trace pairing and equality of dimensions.

Fix a desired pair (c, λ) with $\lambda(x) = 2c$. Choose a K-linear h whose trace functional restricts to $\lambda/2$ on U, and define

$$b_h(u, w) = \varphi(u)h(w) + h(u)\varphi(w) - h(x)\varphi(u)\varphi(w).$$

This symmetric K-bilinear form vanishes on $\widehat{\mathcal{O}} \times \widehat{\mathcal{O}}$ and satisfies $b_h(x, u) = h(u)$. Hence

$$p_h(u) := b_h(u, u), \qquad Q_h(u) := \mathrm{Tr}_{K/\mathbb{F}_q}(\gamma p_h(u))$$

is a pull-back UOV form with

$$Q_h(x) = \mathrm{Tr}(\gamma h(x)) = c, \qquad d(Q_h)_x(u) = 2\,\mathrm{Tr}(\gamma h(u)) = \lambda(u) \quad (u \in U).$$

This proves the first two claims.

For the last claim, compare coefficients in central coordinates. If the two evaluation functionals are proportional by $\rho \in \mathbb{F}_q^*$, nondegeneracy of the trace pairing gives

$$v_z v_z^{\mathsf{T}} = \rho v_x v_x^{\mathsf{T}}, \qquad v_z o_z^{\mathsf{T}} = \rho v_x o_x^{\mathsf{T}}.$$

The first rank-one identity gives $v_z = \lambda v_x$ and $\lambda^2 = \rho$ for some $\lambda \in K^*$. The second then gives $o_z = \lambda o_x$, so $z = \lambda x$ and $\lambda^2 \in \mathbb{F}_q$. Finally,

$$\gcd(2(q-1), q^3 - 1) = q - 1$$

because $q^2 + q + 1$ is odd. Thus $\lambda^{q-1} = 1$ and $\lambda \in \mathbb{F}_q$. □

4.2 Normalize every target coordinate except one

If $y = 0$, then $x = 0$ is already a valid preimage, so only the nonzero case needs analysis. Let $y \in \mathbb{F}_q^m$ be nonzero. Choose a public matrix $T_y \in \mathrm{GL}_m(\mathbb{F}_q)$ whose first $m - 1$ rows span $y^{\perp}$ and whose last row has inner product one with y. Then

$$T_y y = (0, \ldots, 0, 1)^{\mathsf{T}}. \tag{14}$$

Write $Q_1, \ldots, Q_m$ for the corresponding output recombinations of the public forms. Since T_y is fixed by y and invertible, product-uniform public forms remain product-uniform in the public-random expanded-key model.

Only Q_1, Q_2, Q_3 are used to construct the attack subspace. Choose a uniformly random seven-dimensional subspace $S_0 \subset E$, restrict these three forms to S_0, and exhaustively test the points of $\mathbb{P}(S_0) \simeq \mathbb{P}^6(\mathbb{F}_q)$. Chevalley–Warning guarantees a nonzero common zero because three quadrics have total degree six in seven variables. Denote the first such point by w.

The hidden oil space is not known to the attacker, so the algorithm cannot test whether w is off oil. The random seed subspace makes this unnecessary: with overwhelming probability it contains no nonzero oil vector at all.

Proposition 4.2 (A random seed misses every fixed oil space)**.** Fix an arbitrary m-dimensional subspace $\widehat{\mathcal{O}} \subset \mathbb{F}_q^n$ and choose a uniform seven-dimensional subspace $S_0 \subset \mathbb{F}_q^n$. Then

$$\Pr[S_0 \cap \widehat{\mathcal{O}} \neq \{0\}] \le \epsilon_{\mathrm{seed}} := \frac{q^m - 1}{q - 1} \frac{q^7 - 1}{q^n - 1}. \tag{15}$$

Consequently, except with probability at most ϵ_{seed} over public attack coins, every nonzero common zero found inside S_0 is off oil. For the three official profiles,

$$\log_2 \epsilon_{\mathrm{seed}} \le -1048.291, \qquad -1551.477, \qquad -2096.594. \tag{16}$$

The bound holds for every fixed hidden oil space; it does not average over key generation.

Proof. There are $(q^m - 1)/(q - 1)$ projective oil points. For any fixed projective point $[u]$, a uniform seven-dimensional subspace contains $[u]$ with probability $(q^7 - 1)/(q^n - 1)$. A union bound proves (15). The numerical values substitute the three official pairs (n, m). □

4.3 A deterministic Chevalley–Warning extension

For a quadratic form Q over a field of odd characteristic, write

$$B_Q(u, z) = \frac{Q(u + z) - Q(u) - Q(z)}{2}$$

for its polar bilinear form. A subspace is *totally isotropic* for Q when Q vanishes identically on it.

Theorem 4.3 (Common-isotropic extension from one seed point)**.** Let $Q_1, \dots, Q_s$ be homogeneous quadrics on $\mathbb{F}_q^n$, and let $0 \neq w$ satisfy $Q_i(w) = 0$ for every i. If

$$n \geq (s+1)r + s, \tag{17}$$

then a deterministic finite algorithm constructs an r-dimensional subspace U containing w on which every Q_i vanishes identically.

At each extension step the algorithm uses linear algebra and examines at most

$$\left|\mathbb{P}^{2s}(\mathbb{F}_q)\right| = \frac{q^{2s+1}-1}{q-1} \tag{18}$$

projective points.

Proof. Suppose a j-dimensional common totally isotropic subspace W containing w has already been constructed. Put

$$L_W = \bigcap_{i=1}^{s} W^{\perp_{B_{Q_i}}}.$$

The sj orthogonality conditions give

$$\dim(L_W/W) \geq n - (s+1)j.$$

For $j \leq r-1$, (17) makes this dimension at least $2s+1$. Choose any $(2s+1)$-dimensional subspace S of the quotient. The s induced quadrics on S have total degree $2s < 2s+1$. The Chevalley–Warning theorem says that their number of common zeros is divisible by the characteristic [33, 34]. Since the zero vector is one zero, there is a nonzero zero. Enumerating (18) finds one deterministically.

Lift that zero to $z \in L_W$. Then $Q_i(z) = 0$ and $B_{Q_i}(z, W) = 0$, so $W \oplus \langle z \rangle$ is again common totally isotropic. Iterating reaches dimension r. □

Corollary 4.4 (Target-orthogonal affine reduction)**.** Let $P : \mathbb{F}_q^n \to \mathbb{F}_q^m$ be any homogeneous quadratic map, let $y \neq 0$, and choose $1 \leq s < m$. After an invertible output recombination sends y to the last coordinate, a nonzero common zero of any s of the first $m-1$ transformed quadrics can be extended to an $(m-s)$-dimensional common totally isotropic subspace whenever

$$n \geq (s+1)m - s^2. \tag{19}$$

On that subspace, the affine problem has $m-s$ equations in $m-s$ variables.

Proof. Apply Theorem 4.3 with $r = m-s$. Its condition becomes $n \geq (s+1)(m-s)+s = (s+1)m - s^2$. The selected s forms vanish identically on the constructed subspace and have zero target, leaving the stated affine residual. □

The affine formulation leaves one variable that homogeneity makes redundant. The next corollary removes it exactly.

Corollary 4.5 (Projective target-orthogonal reduction)**.** Assume that q is odd. Under the hypotheses of Theorem 4.4, put $r = m-s$ and let U be the constructed r-dimensional subspace. Normalize the target to $(0, \dots, 0, 1)$ and write

$$H_1, \dots, H_{r-1}, H_0$$

for the restrictions to U of the remaining transformed forms, with H_0 corresponding to the nonzero target coordinate. Then:

1. affine solutions of
$$H_1(z) = \cdots = H_{r-1}(z) = 0, \qquad H_0(z) = 1 \tag{20}$$
modulo the pair $z \sim -z$ are in bijection with projective points $[z] \in \mathbb{P}(U)$ satisfying
$$H_1(z) = \cdots = H_{r-1}(z) = 0 \tag{21}$$
and $H_0(z)$ a nonzero square in $\mathbb{F}_q$;
2. a solution of (20) has invertible affine Jacobian if and only if its projective point is nonsingular for (21); and
3. after choosing a uniform nonzero linear form $L \in U^*$, the chart $L(z) = 1$ turns (21) into $r-1$ quadrics in $r-1$ affine variables. Any fixed projective root lies in this chart with probability
$$1 - \frac{q^{r-1} - 1}{q^r - 1} > 1 - \frac{1}{q}.$$

Thus the square solver core has dimension $m - s - 1$ and geometric degree at most 2^{m-s-1}.

Proof. For a nonzero common zero z of $H_1, \dots, H_{r-1}$, replacing z by λz multiplies every quadratic value by λ^2. Hence some representative satisfies $H_0(\lambda z) = 1$ exactly when $H_0(z)$ is a nonzero square, and then there are exactly two choices λ and $-\lambda$.

Euler's identity gives $dH_i(z)(z) = 2H_i(z) = 0$ for $1 \le i < r$, whereas $dH_0(z)(z) = 2$. The first $r-1$ Jacobian rows therefore have rank $r-1$ exactly when their common kernel is the radial line $\langle z \rangle$; in that case the last row is not in their span and the affine Jacobian has rank r. Conversely, an invertible affine Jacobian forces the first $r-1$ rows to have rank $r-1$. Finally, among the $q^r - 1$ nonzero linear forms on U, exactly $q^{r-1} - 1$ vanish at a fixed projective point. Dehomogenizing in a chart that contains the point preserves nonsingularity and gives the claimed square affine system. □

For the QR-UOV parameters, set $s = 3$ and $r = m - 3$. The subspace condition is
$$n \ge 4m - 9. \tag{22}$$

The three official profiles give
$$(n, 4m - 9) = (210, 207), \quad (306, 303), \quad (411, 411).$$

The resulting projective solver dimensions are
$$k = r - 1 = m - 4 \in \{50, 74, 101\}. \tag{23}$$

The seed root and the extension are both obtained by finite exhaustive searches, so this algebraic reduction is unconditional. One seed search and the $r-1$ extension steps each evaluate three seven-variable quadrics on at most $|\mathbb{P}^6(\mathbb{F}_{127})|$ points. The companion audit additionally overcharges uniform seed generation, recomputation of every polar-orthogonal space by dense elimination, the random chart, and the complete success normalization. The resulting preprocessing bounds are
$$2^{63.545}, \qquad 2^{64.077}, \qquad 2^{64.501}$$

Boolean gates, negligible beside every terminal solver estimate below.

4.4 An eligible nonsingular projective root with constant probability

Let U be the subspace from Theorem 4.3, and choose an $\mathbb{F}_q$-basis to identify U with $\mathbb{F}_q^r$. Under the normalization (14), write

$$H_j = Q_{3+j}|_U \quad (1 \le j \le r-1), \qquad H_0 = Q_m|_U.$$

The affine residual is

$$H_1(z) = \cdots = H_{r-1}(z) = 0, \qquad H_0(z) = 1, \qquad r = m-3. \tag{24}$$

The projective solver uses only the first $r-1$ equations; H_0 filters and scales their projective roots.

The next statement gives an explicit success probability rather than assuming that a random-looking square system has a root. Let

$$\pi_{r-1}(q) = \prod_{j=1}^{r-1}(1-q^{-j}). \tag{25}$$

Proposition 4.6 (Affine nonsingular-root probability)**.** Condition on the selected forms, the attack coins used to construct U, and an off-oil seed point. Put

$$d = \dim_{\mathbb{F}_q}(U \cap \widehat{\mathcal{O}}), \qquad \mu = 1 - q^{d-r}.$$

Then $d \le r-1$. Over the independent remaining public-random QR-UOV forms, the probability that (24) has an $\mathbb{F}_q$-rational root with invertible $r \times r$ Jacobian is at least

$$\frac{\mu\, \pi_{r-1}(q)^2}{2 + \mu - (q-1)q^{-r}}. \tag{26}$$

For $q = 127$ and each of $r = 51, 75, 102$, this is at least

$$p_* = 0.326336998767509\ldots. \tag{27}$$

Proof. The off-oil seed point belongs to U, so U is not contained in $\widehat{\mathcal{O}}$ and $d \le r-1$. Let $S = U \setminus \widehat{\mathcal{O}}$ and let M count all roots of (24) in S. For a fixed $x \in S$, Theorem 4.1 gives

$$\Pr[x \text{ is a root}] = q^{-r}.$$

Hence $\mathbb{E}[M] = |S|q^{-r} = \mu$.

For two off-oil points, the evaluation functionals are independent unless the points are scalar multiples. If $z = \lambda x$ and both solve (24), homogeneity and the last equation give $\lambda^2 = 1$. By Theorem 4.1, the only dependent ordered pairs that can both solve the target are therefore $z = x$ and $z = -x$; the other $q-3$ nontrivial base-field multiples have joint probability zero. Every nonproportional ordered pair has joint probability q^{-2r}. Consequently,

$$\mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}. \tag{28}$$

Let N count only roots with invertible Jacobian. At a fixed root x, the conditional one-jet statement makes the Jacobian rows independent and uniform subject only to Euler's relation $dQ_x(x) = 2Q(x)$. The target vector is nonzero, so one column is fixed and nonzero while the other $r-1$ columns are uniform. Thus

$$\Pr[J(x) \text{ invertible} \mid x \text{ is a root}] = \pi_{r-1}(q),$$

and $\mathbb{E}[N] = \mu\pi_{r-1}(q)$. Since $N^2 \le M^2$, the second-moment inequality gives

$$\Pr[N > 0] \ge \frac{\mathbb{E}[N]^2}{\mathbb{E}[M^2]},$$

which is (26). The right-hand side increases with μ, so use $\mu \ge 1 - q^{-1}$ and substitute the three values of r. $\square$

Corollary 4.7 (Eligible projective-root probability)**.** Under the same conditioning, the probability that the projective system

$$H_1 = \cdots = H_{r-1} = 0 \quad \text{in } \mathbb{P}^{r-1}(\mathbb{F}_q)$$

has a nonsingular point $[z]$ for which $H_0(z)$ is a nonzero square is at least the bound in (26), and hence at least p_* for every official profile.

Proof. By Theorem 4.5, every affine nonsingular root of (24) maps to an eligible nonsingular projective root; the two affine roots z and $-z$ map to the same projective point. Existence of an affine root therefore implies the displayed projective event. □

The factor $1/p_* = 3.0643169600\ldots$ costs 1.615566 bits in the work-to-success ratio. A random chart adds less than $\log_2(127/126) = 0.011405$ bits. Unlike a Poisson heuristic, the root bound follows from the QR-UOV pull-back distribution and a second-moment argument. The seed-disjointness statement is fixed-key; the residual-root statement averages over the independent-graph idealized expansion model. We do not claim that repeated invocations on every fixed seeded key are independent.

4.5 Verified projective direct forgery

Theorem 4.8 (Projective target-orthogonal QR-UOV forgery)**.** Work in the independent-graph public-random model and the official degree-three pull-back. For any nonzero target y, the following public algorithm has one-sided error.

1. Normalize the target to $(0, \ldots, 0, 1)$ as in (14).
2. Choose a uniform seven-dimensional seed subspace, enumerate its projective points, and take a nonzero common zero of the first three forms.
3. Use Theorem 4.3 to construct an $(m-3)$-dimensional common totally isotropic subspace for those forms.
4. Choose a uniform nonzero chart form L on that subspace. Dehomogenize the $m-4$ remaining zero-target quadrics in the chart $L = 1$ and apply Theorem 7.6 to the resulting square system in
$$k = m - 4$$
variables.
5. For each represented $\mathbb{F}_q$-rational root, recover its projective point, test whether the last quadratic value is a nonzero square, scale to make that value one, reverse the public transformations, and evaluate the unchanged public map.

Every returned vector is a valid signature preimage. Over the independent-graph public-random key and the public attack coins, one complete invocation reaches an off-oil residual with a represented eligible $\mathbb{F}_q$-rational nonsingular projective root with probability at least

$$(1 - \epsilon_{\text{seed}})p_* \left(1 - \frac{1}{q}\right) \left(1 - \frac{2^{2k}}{|k'|}\right), \qquad k = m - 4, \tag{29}$$

where $k'/\mathbb{F}_q$ is the solver working extension. Taking

$$[k' : \mathbb{F}_q] = 15, \qquad 22, \qquad 30$$

at Levels I, III, and V gives the finite costs reported in Section 10. The inverse of (29) is used as a conservative work-to-success normalization; it is not asserted to be a fixed-key Las Vegas restart factor.

Proof. The first $m-1$ target coordinates are zero. By Theorem 4.2, the seed subspace is disjoint from the hidden oil space with probability at least $1-\epsilon_{\text{seed}}$ for every fixed key. Chevalley–Warning then supplies an off-oil seed root, and Theorem 4.3 constructs the advertised subspace. Conditional on the selected first three forms, the graph, the attack coins, and this off-oil seed point, the remaining transformed forms are still independent uniform pull-back forms. Theorem 4.7 supplies an eligible nonsingular projective root with probability at least p_*. A random nonzero chart contains a fixed such root with probability greater than $1-1/q$. The finite-characteristic solver includes every root in the chart with the separator probability in (29). Square testing and scaling recover the associated affine solutions exactly. Re-evaluation of all m original public forms rejects every malformed or extraneous candidate, so the algorithm cannot output an invalid signature. □

4.6 The fixed-output security scissors

The direct reduction becomes especially informative when combined with the directional equivalent-key attack. Let the quotient degree be ℓ, so that

$$n = \ell V + m.$$

Applying Theorem 4.4 with a general value of s gives the required $(m-s)$-dimensional common-isotropic subspace whenever

$$\ell V \geq s(m-s). \tag{30}$$

By Theorem 4.5, the corresponding nonzero-target solver core has dimension $m-s-1$.

Theorem 4.9 (Projective security scissors at fixed output count)**.** Work over a field of odd characteristic. Fix m and quotient degree ℓ. For every nonzero target, within the two attack families analyzed in this paper an algebraic core of dimension at most $m-\ell-1$ is available for every integer vinegar count V:

$$r_{\text{attack}}(V) \leq m-\ell-1. \tag{31}$$

More precisely:

1. if $V \leq m-\ell-1$, directional equivalent-key recovery uses a square core of dimension V;
2. if $V \geq m-\ell$, target-orthogonal slicing with $s=\ell$, followed by projectivization, uses a square core of dimension $m-\ell-1$.

These cases cover every integer V. For QR-UOV, $\ell=3$, so the ceiling is $m-4$.

Proof. In the first case, $V<m$, so the m directional equations admit a V-equation square recombination and the planted-root attack of Section 5 has core $V \leq m-\ell-1$. In the second case,

$$\ell V \geq \ell(m-\ell),$$

which is (30) with $s=\ell$. The affine direct residual has dimension $m-\ell$ and Theorem 4.5 removes its radial coordinate, leaving core $m-\ell-1$. Since consecutive integers $m-\ell-1$ and $m-\ell$ meet the two inequalities, no case is omitted. □

The theorem is an algebraic-core statement, not a claim that running time depends only on the number of variables. For the official degree-three sets, Theorems 4.2 and 4.7 supply the success factors charged in the concrete ledger. Its design implication is exact: once V reaches $m-\ell$, increasing V while holding m fixed no longer enlarges the projective direct-forgery core. Conversely, decreasing V into the other case makes directional recovery no larger than the same ceiling. Moving both attack lines therefore requires changing m as well as V, or modifying the construction so that one reduction no longer applies.

4.7 Exact small-parameter execution

The companion program `QRUOV_direct_qr_toy.py` instantiates the degree-three model over $\mathbb{F}_{3^3}$, generates independent extension-field UOV forms, pulls each back by an explicit trace, hides the oil space with a random extension-field change of variables, and performs the public attack entirely over $\mathbb{F}_3$. It now executes the projective reduction itself: after building U, it enumerates the projective roots of the remaining zero-target forms and recovers the affine scale from the final equation. In the frozen 200-key run with

$$(V, O, n, m, r, k) = (3, 2, 15, 6, 3, 2),$$

199 keys yielded a verified forgery within twelve outer trials. Every returned vector passed both the base-field public map and an independent extension-field evaluator; the remaining key produced no candidate within the fixed trial budget. 185 of the 200 final seed subspaces were disjoint from the hidden oil space, while all 200 selected seed points were off oil. The distinction is expected at this tiny field size: seed-space disjointness is a sufficient production-scale guarantee, not a condition that the public algorithm tests. This is a correctness regression test, not a production benchmark.

Remark 4.10 (Relation to Hashimoto and the specification). The affine reduction to $m-3$ variables is prior art. Hashimoto's underdetermined-MQ method and the QR-UOV specification already analyze those residual sizes [30, 2]. The exact projective quotient to $m-4$ variables is additional: it uses full target normalization, solves only the homogeneous zero-target equations modulo scaling, and lets the final nonzero-target equation reconstruct the affine representative. The other upgrades are the finite Chevalley–Warning construction, fixed-key seed bound, explicit degree-three nonsingular-root probability, and finite-characteristic solver cost.

5 Graph-direction and derivative-kernel recovery

The attack has two local endpoints. The graph-direction endpoint is the stronger structural statement: one affine root determines the complete hidden graph by public linear algebra. The derivative-kernel endpoint is retained for the faster projective route. Both endpoints are followed by the same exact graph, pull-back, factorization, and signing checks.

5.1 A public graph direction produces a planted root

Fix a direction $c \in K^O$. Substitute the public oil coordinate c and the negated vinegar coordinate $-x$ into the public form P_i:

$$q_{i,c}(x) := P_i(-x, c) = x^\top A_i x - 2x^\top E_i c + c^\top C_i c, \qquad x \in K^V. \tag{32}$$

The system $q_{1,c} = \cdots = q_{m,c} = 0$ is public and has only V variables. Define the public $m \times V$ matrix

$$\mathcal{D}_c(x) := \begin{pmatrix} (E_1 c - A_1 x)^\top \\ \vdots \\ (E_m c - A_m x)^\top \end{pmatrix}. \tag{33}$$

Recall $B_i = E_i - A_i G$, and write $\mathcal{B}(c)$ for the matrix whose ith row is $(B_i c)^\top$.

Proposition 5.1 (Graph-direction identity). For every $c \in K^O$, the point

$$x_c := Gc$$

is a common zero of the public system (32). More precisely, for every $z \in K^V$,

$$q_{i,c}(Gc + z) = z^\top A_i z - 2(B_i c)^\top z. \tag{34}$$

Consequently,

$$\mathcal{D}_c(x_c) = \mathcal{B}(c), \qquad J_q(x_c) = -2\mathcal{B}(c).$$

Proof. Expand (32) at $x = Gc + z$. Its constant term is

$$c^\top (G^\top A_i G - G^\top E_i - E_i^\top G + C_i)c = 0$$

by (4). Its linear term is $2z^\top (A_i G - E_i)c = -2(B_i c)^\top z$, proving (34). The two matrix identities follow by differentiation and by $E_i - A_i G = B_i$. □

Proposition 5.2 (One regular direction determines the entire graph)**.** Let $c \in K^O$, let x be a common zero of all the equations $q_{i,c}$, and suppose $\operatorname{rank} \mathcal{D}_c(x) = V$. For $d \in K^O$, define

$$b_{c,d}(x)_i := c^\top C_i d - x^\top E_i d \qquad (1 \le i \le m). \tag{35}$$

Then the overdetermined public system

$$\mathcal{D}_c(x) y = b_{c,d}(x) \tag{36}$$

has at most one solution $y \in K^V$. If $x = Gc$, its unique solution is $y = Gd$. Thus solving (36) for the O coordinate vectors $d = e_j$ recovers G exactly. Requiring every identity (4) then gives a public polynomial-time certificate of a valid expanded UOV decomposition.

Proof. Full column rank gives uniqueness. For $x = Gc$, multiply (4) on the left by $c^\top$ and on the right by d. Rearrangement yields

$$(E_i c - A_i G c)^\top G d = c^\top C_i d - (Gc)^\top E_i d,$$

which is exactly row i of (36). Hence Gd is the unique solution. The final identities are precisely the hypotheses of Theorem 3.4. □

Proposition 5.3 (Several partial-rank directions can be combined)**.** Let $c_1, \ldots, c_r \in K^O$ and suppose that roots $x_j = Gc_j$ of the corresponding public direction systems have been found. Stack their public completion matrices and right-hand sides:

$$\mathcal{D}_{\boldsymbol{c}}(\boldsymbol{x}) := \begin{pmatrix} \mathcal{D}_{c_1}(x_1) \\ \vdots \\ \mathcal{D}_{c_r}(x_r) \end{pmatrix}, \qquad b_{\boldsymbol{c},d}(\boldsymbol{x}) := \begin{pmatrix} b_{c_1,d}(x_1) \\ \vdots \\ b_{c_r,d}(x_r) \end{pmatrix}.$$

If the stacked matrix has column rank V, then for every $d \in K^O$ the public system

$$\mathcal{D}_{\boldsymbol{c}}(\boldsymbol{x}) y = b_{\boldsymbol{c},d}(\boldsymbol{x})$$

has the unique solution $y = Gd$. Thus no individual direction needs full rank: enough complementary derivative information across several recovered direction roots also determines the whole graph.

Proof. For each j, the proof of Theorem 5.2 shows that Gd satisfies the jth block of equations. It therefore satisfies the stacked system. Full column rank gives uniqueness. □

Remark 5.4 (A compact attack certificate). A successful graph-direction branch can be certified by the tuple (c, x, G) together with one full-rank minor of $\mathcal{D}_c(x)$. A verifier checks the m equations $q_{i,c}(x) = 0$, the O linear systems (36), all m graph identities, the inverse pull-back, and the signing witness. No claim about the other roots or about completeness of the solver output enters this certificate.

5.2 How often a direction is regular, and how to choose a square core

Put

$$\rho_{m,V}(Q) := \prod_{a=0}^{V-1}(1 - Q^{a-m}), \qquad \tau_{m,V}(Q) := 1 - \rho_{m,V}(Q). \tag{37}$$

Thus $\rho_{m,V}(Q)$ is the full-column-rank probability of a uniform $m \times V$ matrix over K.

The direction system has m equations but only V variables. The obvious deterministic reduction to a square system tries all $\binom{m}{V}$ row subsets. The next lemma replaces that list by an optimal-size family of public linear combinations.

Index the m equations by $r = 0, \ldots, m-1$. For $0 \le a < V$, write $(r)_a = r(r-1)\cdots(r-a+1)$, with $(r)_0 = 1$, and define

$$\mathsf{C}(t)_{a+1,r+1} := (r)_a t^{r-a} \qquad (t \in K), \tag{38}$$

where the entry is zero when $r < a$. The first V columns of $\mathsf{C}(t)$ are triangular with diagonal $0!, 1!, \ldots, (V-1)!$. They are therefore nonsingular because $V - 1 < 127 = \operatorname{char} K$.

Lemma 5.5 (A one-parameter rank condenser)**.** Let $B \in K^{m\times V}$ have column rank V, where $m \le 127$. Then

$$W_B(T) := \det(\mathsf{C}(T)B) \tag{39}$$

is a nonzero polynomial of degree at most $V(m-V)$.

Proof. For column j of B, form the polynomial $f_j(T) = \sum_{r=0}^{m-1} B_{r+1,j}T^r$. The columns of B are independent exactly when the polynomials $f_1, \ldots, f_V$ are independent, and $\mathsf{C}(T)B$ is their ordinary Wronskian matrix.

Choose another basis $g_1, \ldots, g_V$ of their span with distinct increasing degrees $0 \le d_1 < \cdots < d_V < m$. The leading coefficient of the Wronskian of the g_j is, up to the nonzero product of their leading coefficients,

$$\det\big((d_j)_a\big)_{0\le a<V,\,1\le j\le V} = \prod_{i<j}(d_j - d_i).$$

This is nonzero in K because all d_j are distinct and below 127. A change of polynomial basis only multiplies the Wronskian by a nonzero scalar, so W_B is nonzero. Finally,

$$\deg W_B \le \sum_{j=1}^{V} d_j - \frac{V(V-1)}{2} \le V(m-V),$$

where the last bound uses the V largest possible distinct degrees below m. $\square$

Proposition 5.6 (Key-by-key sampling and public-random density)**.** For a fixed expanded key, suppose some $V \times V$ minor of $\mathcal{B}(c)$ is not the zero polynomial in c. A fresh uniform $c \leftarrow K^O$ then satisfies $\operatorname{rank}\mathcal{B}(c) = V$ with probability at least

$$1 - \frac{V}{Q}. \tag{40}$$

Conditional on this event, a fresh uniform $t \leftarrow K$ makes $\mathsf{C}(t)\mathcal{B}(c)$ invertible with probability at least

$$1 - \frac{V(m-V)}{Q}. \tag{41}$$

In the public-random model, $\mathcal{B}(c)$ is a uniform $m \times V$ matrix for every fixed nonzero $c \in K^O$. In particular, one preselected coordinate direction fails with probability $\tau_{m,V}(Q)$, whose base-two logarithm is

$$-62.898161, \qquad -62.898161, \qquad -83.864216$$

at Levels I, III, and V. More strongly, the O coordinate-direction matrices $\mathcal{B}(e_1), \ldots, \mathcal{B}(e_O)$ are independent uniform matrices. Therefore

$$\Pr\bigl[\operatorname{rank}\mathcal{B}(e_j) < V \text{ for every } j\bigr] = \tau_{m,V}(Q)^O, \tag{42}$$

with base-two logarithms

$$-1132.166907, \qquad -1635.352198, \qquad -2935.247544.$$

Proof. Every maximal minor of $\mathcal{B}(c)$ is homogeneous of degree V in c, so Schwartz–Zippel gives (40). Conditional on full rank, Theorem 5.5 and the univariate root bound give (41).

Under Theorem 3.1, each B_i is an independent uniform linear map $K^O \to K^V$. For fixed nonzero c, the vector $B_i c$ is uniform in K^V, independently across i. For the coordinate directions, the vectors $B_i e_j$ use independent columns of the B_i, proving both the fixed-direction law and (42). The displayed values are exact evaluations of the rank formulas. □

Corollary 5.7 (A short deterministic public list)**.** Fix any public list of $V(m-V)+1$ distinct elements of K. Whenever $\mathcal{B}(c)$ has column rank V, at least one t on the list makes the square system

$$\mathsf{C}(t)\,(q_{1,c}, \ldots, q_{m,c})^{\mathsf{T}} = 0 \tag{43}$$

have the planted root as a nonsingular solution.

For one fixed coordinate direction, the list sizes at Levels I, III, and V are

$$105, \qquad 153, \qquad 307.$$

Scanning all coordinate directions uses respectively

$$1{,}890, \qquad 3{,}978, \qquad 10{,}745$$

square systems, and fails in the public-random model only with probability $\tau_{m,V}(Q)^O$. Thus all outer choices can be deterministic; only the polynomial-system solver needs random coins.

Proof. The determinant in (39) has degree at most $V(m-V)$ and cannot vanish at every point of a longer distinct list. At the planted root, the Jacobian of (43) is $-2\mathsf{C}(t)\mathcal{B}(c)$ by Theorem 5.1. The numerical counts use $m - V \in \{2, 2, 3\}$. □

Corollary 5.8 (Category-scale deterministic direction lists)**.** Use the first

$$r_{\mathrm{dir}} = 3, \qquad 4, \qquad 4$$

coordinate directions at Levels I, III, and V, and for each direction use the condenser list from Theorem 5.7. The resulting fully public lists contain

$$315, \qquad 612, \qquad 1228$$

square systems. In the public-random model, the probability that none has the planted point as a nonsingular root is exactly $\tau_{m,V}(Q)^{r_{\mathrm{dir}}}$, with base-two logarithms

$$-188.694, \qquad -251.593, \qquad -335.457. \tag{44}$$

These bounds are already below the Level-I, Level-III, and Level-V category exponents $143, 207, 272$. Scanning all O coordinate directions gives the stronger probabilities in (42).

Proof. The coordinate-direction derivative matrices are independent by Theorem 5.6. If one of the first r_{dir} matrices has full column rank, Theorem 5.7 supplies a condenser value with nonsingular planted Jacobian. Multiplying the one-direction failure logarithm by r_{dir} gives (44); multiplying the per-direction list lengths $105, 153, 307$ gives the displayed system counts. $\square$

Remark 5.9 (The list length is geometrically optimal). The Grassmannian of V-subspaces of an m-dimensional space has dimension $V(m-V)$. For any fixed full-rank linear map $R : K^m \to K^V$, the subspaces on which R is singular form a hyperplane section in Plücker coordinates. Over an algebraic closure, at most $V(m-V)$ such hyperplane sections always have a common point. Hence no family of at most $V(m-V)$ linear recombinations can work for every full-rank $m \times V$ matrix after all field extensions. The family in Theorem 5.7 meets this natural lower bound exactly.

Randomized versus deterministic use. Sampling one field element t is convenient for the expected-cost theorem. The short list in Theorem 5.7 removes that choice entirely and is far smaller than enumerating all V-row subsets.

5.3 Derivative-kernel recovery for the projective route

For a public common zero x, form

$$C_x = (P_1 x \ \cdots \ P_m x) \in L^{(V+O)\times m}.$$

Lemma 5.10 (Derivative-kernel recovery)**.** If $x \in \mathbb{P}(\mathcal{O}_L)$, then $\mathcal{O}_L \subseteq \ker C_x^{\mathsf{T}}$. If $\operatorname{rank} C_x = V$, equality holds.

Proof. For $u \in \mathcal{O}_L$, (1) gives $u^{\mathsf{T}} P_i x = 0$ for every i. If $\operatorname{rank} C_x = V$, its transpose kernel has dimension $(V+O) - V = O$. $\square$

At the oil point, a smooth restricted core has derivative rank V. Total isotropy bounds the ambient derivative rank by the same value, so Theorem 5.10 recovers the complete oil space. Row reduction into the public graph chart, Frobenius descent to K, verification of (4), and Theorem 3.4 then recover the expanded decomposition. Exact factorization certifies that decomposition, while Theorem 3.7 supplies an invertible oil matrix whose inverse certifies target inversion.

Optional identification with the planted oil space. Neither endpoint needs uniqueness: the algorithm keeps every exactly verified candidate and tests all of them for a signing witness. Section D separately shows that a second K-rational oil space is extraordinarily unlikely. Outside that additional event, every accepted oil space is the planted one.

6 Jacobian localization removes global regularity

Fix $c \in K^O$ and a public recombination $R \in K^{V\times m}$. Let

$$f = (f_1, \ldots, f_V)^{\mathsf{T}} := R(q_{1,c}, \ldots, q_{m,c})^{\mathsf{T}}$$

and put

$$h(x) := \det J_f(x). \tag{45}$$

The graph-direction condition is local: if $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, then Theorem 5.1 gives

$$h(Gc) = (-2)^V \det(R\mathcal{B}(c)) \neq 0. \tag{46}$$

No assertion is needed about the system on the complementary hypersurface $h = 0$.

Lemma 6.1 (Localization gives a reduced regular sequence)**.** Let k be a perfect field, let $f_1, \ldots, f_V \in k[x_1, \ldots, x_V]$, and let $h = \det J_f$. In the localization

$$S_h := k[x_1, \ldots, x_V]_h,$$

the sequence $f_1, \ldots, f_V$ is regular. For every $1 \leq s \leq V$, the ideal $(f_1, \ldots, f_s)S_h$ is radical. Equivalently, the sequence is reduced and regular on the open set $h \neq 0$.

Proof. Let $\mathfrak{p}$ be a prime of S_h containing the first s equations. Because h is a unit in S_h, the full Jacobian matrix is invertible modulo $\mathfrak{p}$. Its first s rows are therefore linearly independent. The Jacobian criterion over a perfect field shows that the local zero set of $f_1, \ldots, f_s$ is smooth of codimension s at every such prime. Hence the localized quotient is reduced and every minimal prime has height s. The ring S_h is regular, hence Cohen–Macaulay; an ideal generated by s elements and of height s is a complete intersection. Thus the generators form a regular sequence, and smoothness gives radicality for every prefix. □

Corollary 6.2 (Only the planted simple root is needed)**.** If $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, then Gc belongs to the locally closed set

$$\{f_1 = \cdots = f_V = 0,\ h \neq 0\}.$$

The input tuple $(f_1, \ldots, f_V, h)$ satisfies hypotheses (H1)–(H2) of the locally closed Kronecker algorithm of Giménez et al. These conclusions are independent of every root and irreducible component contained in $h = 0$.

Proof. Equation (46) puts the planted root in the open set. Theorem 6.1 is exactly the reduced-regular-sequence condition required after localization. □

Remark 6.3 (Why this is a separate geometric route). The local theorem does not use the projective slice, the off-oil incidence, the cumulative-degree bound, the joint discriminant, smoothness of the full projective core, or radicality of components on $h = 0$. A failure in any of those arguments can affect the faster projective line without affecting graph-direction recovery.

7 Finite-characteristic nonsingular-root recovery

The directional reduction does not need a description of every component of the square core. It needs one isolated root at which the Jacobian is invertible. This section adapts the symbolic-homotopy algorithm of Safey El Din and Schost to that exact task in characteristic 127.

7.1 The solver target and its degree parameters

Let k be a perfect field and let

$$f = (f_1, \ldots, f_V) \in k[X_1, \ldots, X_V]^V$$

be a square system of affine quadrics. Write

$$Z_{\text{ns}}(f) := \{x \in \overline{k}^V : f(x) = 0,\ \det J_f(x) \neq 0\}. \tag{47}$$

Every point in $Z_{\text{ns}}(f)$ is isolated. Bézout's theorem gives

$$C := 2^V, \qquad C_0 := 2^V + V2^{V-1} = 2^{V-1}(V + 2). \tag{48}$$

Here C bounds the number of isolated roots, while C_0 bounds the degree of the one-parameter homotopy curve. Rational reconstruction of a degree-C_0 curve uses coefficients through precision $2C_0$; the lifting ledger below therefore runs to that precision. The second formula is the coefficient sum of $(\vartheta_0 + 2\vartheta)^V$ after discarding ϑ_0^2 and ϑ^{V+1}, exactly as in the multihomogeneous degree definition of Safey El Din and Schost [27].

The published algorithm starts from a product system with known simple roots, follows its C branches symbolically, specializes at the target fiber, and removes singular outputs by a Jacobian-based CLEAN operation. Its large-characteristic hypothesis supports the particular finite families used for the start and the separator. We replace both families.

7.2 A simple product start over any sufficiently large finite field

Lemma 7.1 (Vandermonde product start)**.** Let k' contain $2V$ distinct elements $a_1, \dots, a_{2V}$. For $a \in k'$, put

$$L_a(X) := X_1 + aX_2 + \cdots + a^{V-1}X_V + a^V, \tag{49}$$

and, for $1 \le i \le V$, set

$$g_i(X) := L_{a_{2i-1}}(X)L_{a_{2i}}(X). \tag{50}$$

Then $g = (g_1, \dots, g_V)$ has exactly $C = 2^V$ affine roots over k'. Every root is simple. A zero-dimensional parametrization of all roots can be constructed in $O(V^2C)$ operations in k'.

Proof. Choose one node $b_i \in \{a_{2i-1}, a_{2i}\}$ from each pair. The equations $L_{b_1} = \cdots = L_{b_V} = 0$ have coefficient matrix

$$\begin{pmatrix} 1 & b_1 & \cdots & b_1^{V-1} \\ \vdots & \vdots & & \vdots \\ 1 & b_V & \cdots & b_V^{V-1} \end{pmatrix},$$

which is Vandermonde and invertible. Hence the choice gives one solution.

At that solution, the monic polynomial

$$T^V + X_V T^{V-1} + \cdots + X_2 T + X_1$$

has $b_1, \dots, b_V$ as its roots. It therefore vanishes at none of the V unchosen nodes. Consequently the solution satisfies exactly one factor in each product g_i, and different choices give different solutions. Conversely, any common zero of the products must choose at least one factor from each pair. It cannot satisfy both factors in one pair, because that would give at least $V + 1$ distinct roots of the displayed monic polynomial. Thus there are exactly 2^V roots.

At a root, row i of the Jacobian of g is the Vandermonde row belonging to the chosen node, multiplied by the nonzero value of the unchosen factor. The Jacobian is therefore an invertible Vandermonde matrix times an invertible diagonal matrix. Enumerating the 2^V choices and forming the corresponding degree-V polynomials gives the stated $O(V^2C)$ bound. □

This construction uses no division by an integer and no condition on the characteristic beyond the existence of $2V$ distinct field elements. For the QR-UOV working extensions below, this condition is trivial.

7.3 A full random separator

A primitive linear form must separate finitely many branch values and avoid finitely many leading-coefficient vectors. The published proof draws from a short integer-indexed family; that is the source of its second characteristic bound. Sampling the entire dual space is simpler over a finite extension.

Lemma 7.2 (Separator over a finite extension)**.** Let A be a domain containing the finite field k', and let $z_1, \dots, z_s \in A^V$ be nonzero. For a uniform $\lambda = (\lambda_1, \dots, \lambda_V) \leftarrow (k')^V$,

$$\Pr[\lambda \cdot z_j = 0 \text{ for some } j] \leq \frac{s}{|k'|}. \tag{51}$$

In particular, if $s \leq C^2$, the separator succeeds with probability at least $1 - C^2/|k'|$.

Proof. Fix a nonzero vector z. Choose an index h with $z_h \neq 0$. After the other $V-1$ coefficients of λ are fixed, at most one value of $\lambda_h \in k'$ can make $\lambda \cdot z = 0$: two such values would give a nonzero scalar in k' annihilating the nonzero domain element z_h. Thus one vector is killed with probability at most $1/|k'|$. A union bound proves the claim. □

Safey El Din and Schost identify at most C^2 vectors that a well-separating form must avoid: differences between start roots, differences between relevant target specializations, and leading-coefficient vectors of branches [27, Lem. 14 and Rem. 15]. Theorem 7.2 therefore replaces their integer family without changing the success criterion.

7.4 The lifting identities do not require large characteristic

For completeness, we isolate the two local identities used by precision doubling. They work over a commutative ring and do not divide by factorials or by the characteristic.

Lemma 7.3 (Newton–Hensel precision doubling)**.** Let R be a commutative ring, let $I \subset R$ be an ideal, and let $F \in R[X_1, \dots, X_V]^V$. Suppose

$$F(x) \equiv 0 \pmod{I^m}$$

and that $J_F(x)$ is invertible modulo I^m. Put

$$x^+ := x - J_F(x)^{-1} F(x) \pmod{I^{2m}}. \tag{52}$$

Then $F(x^+) \equiv 0 \pmod{I^{2m}}$.

Proof. The correction $h = x^+ - x$ lies in I^m. Polynomial Taylor expansion gives

$$F(x+h) = F(x) + J_F(x)h + R_2(x,h),$$

where every monomial in R_2 contains at least two coordinates of h. Hence $R_2(x,h) \in I^{2m}$. The first two terms cancel by (52). No denominator is used. □

Lemma 7.4 (First-order change of primitive parameter)**.** Work modulo T^{2m}. Let $Q(T,U)$ be monic in U, let $X(T,U) \in (R[T,U]/(T^{2m},Q))^V$, and let λ be linear. Suppose

$$e(T,U) := \lambda(X(T,U)) - U \in T^m R[T,U]/(T^{2m},Q).$$

Define

$$X^* := X - (\partial_U X)e, \qquad Q^* := Q - (\partial_U Q)e. \tag{53}$$

Then the change of parameter $U_{\text{old}} = U - e(T,U)$ represents the same lifted branches modulo T^{2m}, and

$$\lambda(X^*) \equiv U \pmod{(T^{2m}, Q^*)}. \tag{54}$$

Proof. Because $e \in T^m$, substitution of $U - e$ into any polynomial is, modulo T^{2m}, its first-order Taylor expansion. This gives exactly (53). Moreover

$$\lambda(X^*) - U = e - (\partial_U \lambda(X))e = -(\partial_U e)e.$$

Differentiation in U does not change T-adic order, so $\partial_U e \in T^m$ and the product lies in T^{2m}. Again, the argument uses no integer denominator. □

7.5 Proof crosswalk for the finite-characteristic solver

Proposition 7.5 (Where the published characteristic bound is used)**.** For a square system of quadrics over a finite perfect field, replace the start construction in Safey El Din–Schost Lemma 7 by Theorem 7.1, and replace the separator family in their Lemma 14 by Theorem 7.2. The remaining correctness proof of their NonsingularSolutions algorithm applies over the extension field without a further lower bound on the characteristic.

Proof. We follow the proof in the order in which its ingredients are used.

Start fiber. The original lower bound $\operatorname{char} k \geq \sum_i d_i$ ensures that its integer-indexed affine forms are distinct and that the product start has the full multihomogeneous root count with invertible Jacobian. Theorem 7.1 supplies the same three facts directly: degree two, exactly C explicit roots, and a simple start fiber.

Homotopy curve. The homotopy is

$$H(T, X) = (1 - T)g(X) + Tf(X).$$

The published construction saturates by the X-Jacobian determinant. Lazard's lemma and the multihomogeneous degree bound then give a one-dimensional curve of degree at most C_0. These arguments require the simple start just established, not an ordering of integers inside the field.

Target specialization. The target system can have fewer than C isolated nonsingular roots and may have singular or positive-dimensional components. The published proof handles this by generalized power series, explicitly because ordinary Puiseux series are not sufficient in arbitrary characteristic [27, Sec. 3.4]. Its valuation and specialization lemmas therefore already cover the finite-field setting.

Precision doubling and primitive normalization. The global lifting step is Newton–Hensel lifting of a geometric resolution. Its local identities are Theorems 7.3 and 7.4; interpolation and polynomial arithmetic use inverses guaranteed by the simple fiber and separating form. No factorial or characteristic-dependent scalar is introduced.

Separator and cleaning. The published well-separating condition is avoidance of at most C^2 nonzero vectors. Theorem 7.2 supplies it with probability at least $1 - C^2/|k'|$. Finally, the published Clean call removes every point where the target Jacobian vanishes. As noted in its Remark 15, every output is therefore a subset of the nonsingular target roots; a bad separator may omit roots or cause failure, but cannot introduce an invalid point.

These replacements cover the two uses of the integer-indexed finite families. All other steps are field algebra, generalized-power-series specialization, or Jacobian cleaning over the perfect extension field. $\square$

Theorem 7.6 (Nonsingular roots in characteristic 127)**.** Let k be a finite field and let $f = (f_1, \ldots, f_V)$ be affine quadrics over k, represented by a straight-line program of size L. Let k'/k be a finite extension containing at least $2V$ elements. There is a randomized algorithm with the following properties.

1. Every returned point belongs to $Z_{\mathrm{ns}}(f)$.
2. Every point of $Z_{\mathrm{ns}}(f)$ is represented in the output with probability at least

$$1 - \frac{C^2}{|k'|}. \tag{55}$$

3. The arithmetic cost is

$$\widetilde{O}\Big(CC_0(L + 2V + V^2)V\Big) \tag{56}$$

operations in k'.

In particular, $|k'| \geq 8C^2$ gives one-invocation inclusion probability at least 7/8.

Proof. Run the symbolic-homotopy algorithm of Safey El Din and Schost with the start from Theorem 7.1 and a uniform full separator from Theorem 7.2. Theorem 7.5 gives correctness over k'. The well-separating event has probability at least (55); on that event the published specialization recovers every isolated nonsingular target branch. The final CLEAN step proves item 1 on every branch, including branches with a bad separator.

For V quadrics in one block, the multihomogeneous quantities in the published Proposition 5 are exactly C and C_0 from (48); the sum of degrees is $2V$. The new start costs $O(V^2C)$ and does not change the dominant lifting term. Substitution into that proposition gives (56). $\square$

Remark 7.7 (One-sided failure is enough for cryptanalysis). The original solver paper notes that it cannot cheaply certify completeness of a generic output. QR-UOV does not need such a certificate. We impose every unrecombined directional equation, require a K-rational root, complete the graph, and verify the unchanged public map. An omitted planted root causes a restart. An extraneous or malformed candidate is rejected.

7.6 End-to-end verified directional recovery

For a random $V \times m$ matrix R over K, put

$$f_{c,R} := R(q_{1,c}, \ldots, q_{m,c})^{\mathsf{T}}.$$

If $\mathcal{B}(c)$ has rank V, then $R\mathcal{B}(c)$ is a uniform $V \times V$ matrix. Write

$$\mu_V(Q) := \prod_{j=1}^{V} (1 - Q^{-j}), \qquad Q = |K| = 127^3. \tag{57}$$

Choose $K' = \mathbb{F}_{Q^r}$ with

$$r := \min\{s \geq 1 : Q^s \geq 8 \cdot 2^{2V}\}. \tag{58}$$

The values are

$$r = 6, \qquad 8, \qquad 10 \tag{59}$$

at Levels I, III, and V.

Theorem 7.8 (Verified directional equivalent-key recovery)**.** Let a fixed expanded QR-UOV public key have a valid graph decomposition G. Suppose

1. some maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c; and
2. the recovered planted decomposition has a nonzero signing determinant polynomial Δ from Theorem 3.7.

Then the following Las Vegas algorithm returns a publicly verified functional equivalent key.

1. Sample $c \leftarrow K^O$ and $R \leftarrow K^{V \times m}$.
2. Run Theorem 7.6 on $f_{c,R}$ over K' from (58).
3. Intersect the returned parametrization with all m original directional equations and with the K-rational locus.
4. For every retained root x, run the graph completion of Theorem 5.2; require all graph identities, the inverse pull-back, exact public-map factorization, and a signing witness.

5. Return the first accepted key; otherwise restart with fresh public coins.

One invocation includes the planted branch with probability at least

$$p_{\mathrm{dir}} \geq \left(1 - \frac{V}{Q}\right) \mu_V(Q) \left(1 - \frac{2^{2V}}{Q^r}\right). \tag{60}$$

The expected numbers of full solver invocations are at most

$$1.0000262, \qquad 1.0000561, \qquad 1.0202201. \tag{61}$$

Every returned key is correct, regardless of whether failed solver invocations were complete.

Proof. By Theorem 5.6, a random direction has full planted derivative with probability at least $1 - V/Q$. Conditional on that event, $R\mathcal{B}(c)$ is uniform square, so it is invertible with probability $\mu_V(Q)$. The planted point is then an isolated nonsingular root of $f_{c,R}$ by Theorem 5.1. Theorem 7.6 includes it with probability at least $1 - 2^{2V}/Q^r$.

The all-equation and rational-root filters retain the planted point. Theorem 5.2 reconstructs G, and Theorems 3.4 and 3.7 supply the exact factorization and signing certificate. Conversely, any candidate that fails an original equation, completion equation, graph identity, pull-back identity, factorization check, or oil-matrix test is rejected. This proves zero-error acceptance. Fresh restarts and (60) give the geometric tail and the numerical expectations. □

Corollary 7.9 (Public-random key density)**.** In the public-random expanded-key model, the probability that every coordinate direction has deficient planted derivative is $\tau_{m,V}(Q)^O$, with base-two logarithms

$$-1132.166907, \qquad -1635.352198, \qquad -2935.247544.$$

The probability that the deterministic pair-line signing search fails has base-two logarithm below

$$-544.820, \qquad -796.276, \qquad -1068.687.$$

Outside their union, a deterministic scan of coordinate directions and signing witnesses contains a successful outer branch; only the internal nonsingular-root solver requires random coins.

Proof. Combine Theorems 3.7 and 5.6. The two exceptional events need not be independent; a union bound suffices. □

8 Concrete solver refinements and the Level-I estimate

The generic root solver does not by itself fix the concrete cost. This section first states the resulting candidate bound. It then gives the arithmetic choices and branch filters used by the final ledger: batched dense lifting, sparse working fields, local separation, and exact Frobenius descent. The section ends with the root-success bound and the finite Level-I ledger.

8.1 Candidate local-separator bound

Let $q = 127$, $a = m - 4$, $C = 2^a$, and $C_0 = 2^{a-1}(a + 2)$ as in the parent attack. The supplied audit optimizes the extension degree over certified sparse irreducible polynomials.

Theorem 8.1 (Candidate local-separator bound)**.** Assume the public-random model, homotopy degree bound, and Boolean-gate conventions of the preceding direct-attack construction and the split-before-lift refinement developed here. Replace global separation by the local conditions in Section 8.5, accept only simple linear target factors, and use the exact descent test in Section 8.6. Then the retry-aware diagnostic costs are those in Table 1. In particular, Level I costs

$$2^{140.383995063}$$

Boolean gates, leaving 2.616004937 bits below the category target.

Table 1: Audited stress estimates. "Global split" is the preceding split-before-lift bundle.

Level	Parent	Global split	Local split	Target	Margin
I	151.639	141.126	140.384	143	+2.616
III	203.507	191.630	191.027	207	+15.973
V	260.389	247.637	247.125	272	+24.875

The Level-I optimum in the main ledger uses $K' = \mathbb{F}_{127^8}$ with the Rabin-certified modulus

$$X^8 + X^4 - 1.$$

Degree 9, using $X^9 + X^2 + 1$, gives total 140.418965 and is retained as a nearby cross-check.

8.2 Dense-quadratic batching and flattened lifting

8.2.1 Why the direct straight-line-program bound is not the last word

For dense quadrics, a direct straight-line program has size $L = O(V^3)$. Substituting it into (56) gives

$$\widetilde{O}(V^5 4^V) \tag{62}$$

operations in K'. This is a valid generic specialization, but it repeatedly charges contractions that all use the same dense quadratic coefficient tensor.

Write

$$f_i(X) = X^\top A_i X + b_i^\top X + c_i. \tag{63}$$

At precision k, the lifted coordinates live in a coefficient algebra

$$\mathcal{A}_k := K'[T, U]/(T^k, Q_k(T, U)), \tag{64}$$

where Q_k is monic of U-degree at most C. The V coordinate vectors form a $V \times (Ck)$ coefficient matrix. Flattening the matrices A_i turns all Jacobian contractions into one rectangular matrix product over K'. Quadratic values follow from the Jacobian and X, while the Newton solve uses matrix algebra over $\mathcal{A}_k$.

Proposition 8.2 (Dense-quadratic lifting ledger)**.** Let ω be an admissible square-matrix multiplication exponent, and let $\mu(C, k)$ be the cost of one multiplication in $\mathcal{A}_k$. The precision-k work of the global Newton lift can be organized using

$$\widetilde{O}(V^3 Ck + (V^\omega + V^2)\mu(C, k)) \tag{65}$$

operations in K'. Summed over precision doubling through $2C_0$, this gives

$$\widetilde{O}(V^3 CC_0 + (V^\omega + V^2)\mu(C, 2C_0)). \tag{66}$$

Proof. Stack the V matrices A_i into a $V^2 \times V$ scalar matrix. Multiplication by the $V \times (Ck)$ lifted-coordinate matrix forms every linear Jacobian contraction simultaneously in $O(V^3Ck)$ scalar operations. This term is written explicitly rather than hidden inside the coefficient-algebra multiplication count.

The identity

$$f_i(X) = \frac{1}{2}X^\mathsf{T}(2A_iX + b_i) + \frac{1}{2}b_i^\mathsf{T}X + c_i$$

computes all values with $O(V^2)$ products in $\mathcal{A}_k$. Newton matrix inversion or solving costs $O(V^\omega)$ such products, and primitive-element renormalization lies within the same envelope. The terminal precision is $2C_0$; quasi-linear multiplication makes the last doubling stage dominate up to logarithmic factors. □

8.2.2 One multiplication, not a product of two multiplication costs

The earlier stress analysis bounded multiplication in (64) by applying a quasi-linear polynomial-multiplication bound first in U and then in T. This gives a product $\mathsf{M}(C)\mathsf{M}(k)$. A bivariate array can instead be flattened once.

Lemma 8.3 (Collision-free Kronecker flattening)**.** Let $R_k = K'[T]/(T^k)$ and let $Q_k(U) \in R_k[U]$ be monic of degree C. After precomputing the reciprocal of the reversed Q_k, one multiplication in $R_k[U]/(Q_k)$ can be performed using at most three ordinary polynomial multiplications whose input length is below $2Ck$, plus $O(Ck)$ additions. The reciprocal precomputation costs $O(\mathsf{M}(2Ck)\log C)$ operations in K'.

Proof. For a polynomial

$$A(U,T) = \sum_{0\le i<C,\ 0\le j<k} a_{i,j}U^iT^j,$$

choose the radix $B = 2C - 1$ and encode

$$U^iT^j \longmapsto Z^{i+Bj}. \tag{67}$$

In a product, the U-exponent is at most $2C - 2 = B - 1$. It therefore never carries into the T digit. The largest input exponent is $(C-1) + B(k-1) < Bk < 2Ck$, so one ordinary univariate multiplication recovers the complete bivariate product modulo T^k.

To reduce the U-degree modulo the monic polynomial Q_k, use the standard reciprocal division algorithm over the coefficient ring R_k. With the reciprocal precomputed, one product obtains the quotient estimate and one product forms the quotient times Q_k. Together with the unreduced product, this is three products of the same padded length. Newton inversion of the reversed monic polynomial gives the stated one-time reciprocal cost over any commutative coefficient ring. □

Corollary 8.4 (Flattened lifting complexity)**.** Let $\mathsf{M}(n)$ denote a quasi-linear polynomial-multiplication cost. The batched lift can be implemented in

$$\widetilde{O}(V^3CC_0 + (V^\omega + V^2)\mathsf{M}(CC_0)) = \widetilde{O}(V^{\omega+1}4^V) \tag{68}$$

operations in K'.

For the explicit stress function

$$\mathsf{M}(n) := n\log_2 n \log_2\log_2 n, \tag{69}$$

the entire precision-doubling schedule for one coefficient-algebra product is bounded by

$$3\mathsf{M}(8CC_0). \tag{70}$$

Proof. Apply Theorem 8.3 at precisions $2C_0, C_0, C_0/2, \ldots$. At the terminal stage, each encoded input has length below $4CC_0$. Monotonicity of the logarithmic factors gives

$$\sum_{j\geq 0} \mathsf{M}\left(\frac{4CC_0}{2^j}\right) < 2\mathsf{M}(4CC_0) < \mathsf{M}(8CC_0).$$

There are three ordinary products per modular multiplication. Reciprocal precomputation contributes an additional logarithmic factor but is dominated by the explicit V^2 outer allowance. Finally, $CC_0 = 2^{2V-1}(V+2)$. □

Remark 8.5 (What was improved and what was corrected). Flattening does not alter the exponential factor 4^V; it replaces the product of two independent quasi-linear overheads by the overhead at the total coefficient-array length. The factor 8 in (70) is important: an earlier draft stopped at precision C_0, whereas rational reconstruction requires $2C_0$. All finite values in Section 10 use the corrected precision.

8.3 Sparse working-field models

Both solvers may choose any convenient representation of their finite working field. The directional solver starts over $K = \mathbb{F}_{127^3}$, so its auxiliary fields must contain K; the direct solver starts over $\mathbb{F}_{127}$ and has no such divisibility requirement. Projectivization lowers the Level-I direct working degree from 16 to 15.

Lemma 8.6 (Explicit sparse auxiliary fields)**.** The following polynomials are irreducible over $\mathbb{F}_{127}$:

$$\phi_{15}(X) = X^{15} + X^{12} - X + 1, \qquad \phi_{18}(X) = X^{18} + X^4 + 1,$$
$$\phi_{22}(X) = X^{22} + X + 1, \qquad \phi_{24}(X) = X^{24} + X^5 - X + 1,$$
$$\phi_{30}(X) = X^{30} + X^7 + 1.$$

The projective direct-forgery solver uses degrees $15, 22, 30$ at Levels I, III, and V. The directional solver uses degrees $18, 24, 30$.

Proof. The companion audit applies Rabin's irreducibility criterion [26]. For every displayed degree d, it verifies

$$X^{127^d} = X \pmod{\phi_d}$$

and, for each prime divisor e of d,

$$\gcd\left(\phi_d, X^{127^{d/e}} - X\right) = 1.$$

All computations are exact over $\mathbb{F}_{127}$, and the machine-readable output records every gcd degree. The same audit certifies the additional sparse fields used in the retuning table of Table 7, including $X^{26} + X^3 - X + 1$. □

Schoolbook convolution is unnecessary at these modest extension degrees. The next statement freezes an explicit recursive circuit rather than invoking an asymptotic multiplication theorem.

Proposition 8.7 (Audited Karatsuba extension-multiplication envelope)**.** Let

$$g_{\mathrm{mul}} = \mathsf{G}_{\mathrm{mul}}(\log_2 127), \qquad g_{\mathrm{add}} = 4\log_2 127.$$

For length-d coefficient vectors, let (M_d, A_d) be chosen recursively as follows. The schoolbook candidate is

$$(d^2, (d-1)^2).$$

For $d \geq 2$, put $a = \lceil d/2 \rceil$ and $b = \lfloor d/2 \rfloor$. The Karatsuba candidate is

$$\bigl(2M_a + M_b,\ 2A_a + A_b + 4d - 4\bigr). \tag{71}$$

Choose whichever candidate has smaller value $Mg_{\mathrm{mul}} + Ag_{\mathrm{add}}$. If s_d is the number of nonleading terms of ϕ_d, one multiplication in $\mathbb{F}_{127^d}$ followed by sparse reduction is then charged by

$$\mathsf{G}^{\mathrm{sparse}}_{127^d} := M_d g_{\mathrm{mul}} + \bigl(A_d + s_d(d-1)\bigr)g_{\mathrm{add}}. \tag{72}$$

For the official working fields, the selected plans are:

d	M_d	A_d	s_d
15	81	332	3
18	153	404	2
22	225	480	2
24	243	512	3
30	243	1112	2

Proof. The schoolbook counts are exact: d^2 coefficient products and $d^2 - (2d-1) = (d-1)^2$ additions. For the recursive candidate, write each operand as a low block of length a plus X^a times a high block of length b. The products of the two low blocks and of their sums have length a; the high-block product has length b, giving $2M_a + M_b$ coefficient multiplications. Forming the sums, subtracting the low and high products from the middle product, and combining the three shifted outputs uses at most $4d-4$ additional coefficient additions, giving (71). Recursion therefore realizes the displayed pair (M_d, A_d).

The unreduced product has only $d-1$ high coefficients. Eliminating each one modulo ϕ_d uses s_d additions or subtractions and no nontrivial constant multiplication because every nonleading coefficient is $+1$ or -1. This proves (72). The companion audit executes the same recursive plan and compares 48 random products against schoolbook convolution before certifying the gate counts. $\square$

8.4 Characteristic polynomials without global separation

Let $\Gamma \subset \mathbb{A}^{1+a}$ be the union of the saturated homotopy-curve components that dominate the t-line, with degree at most $D = C_0$. The Jacobian saturation removes the non-etale locus, so over $K'(t)$ its generic fiber is a finite product of separable field extensions,

$$B = K'(t)[X_1, \ldots, X_a]/J, \qquad \dim_{K'(t)} B = C.$$

Write x_j for the image of X_j and M_{x_j} for multiplication by x_j on B.

Introduce independent coefficients $\Lambda = (\Lambda_1, \ldots, \Lambda_a)$ and the generic linear element

$$u_\Lambda = \sum_j \Lambda_j x_j.$$

Its characteristic polynomial is

$$\chi_\Lambda(t, U) = \det\left(UI - \sum_j \Lambda_j M_{x_j}\right). \tag{73}$$

Schost proves two facts that are exactly suited to the present use: specialization of Λ gives the characteristic polynomial of the specialized linear element, and the generic characteristic polynomial can be written

$$\chi_\Lambda = \frac{\Xi_\Lambda}{\Theta},$$

where the t-degrees of Ξ_Λ and Θ are at most $\deg\Gamma \le D$ and Θ is independent of Λ [28, Lemmas 2–3 and Proposition 2]. This is the characteristic-polynomial, or generic u-resultant, form of the parametric degree theorem.

Schost phrases one corollary using a primitive element with coefficients in the ground field, together with a field-size hypothesis. The generic characteristic polynomial itself does not require such an element. After extending to an algebraic closure of $K'(t)$, the finite etale algebra has C distinct geometric points. For every distinct pair, the difference

$$\Lambda \cdot (x_s - x_r)$$

is a nonzero polynomial in the independent indeterminates Λ. Hence u_Λ separates the geometric points over the rational-function field $K'(t)(\Lambda)$ and is primitive there, irrespective of the size of K'. Apply Schost's degree argument after this transcendental scalar extension. The determinant in (73) is formed from the original multiplication matrices, so its coefficients already lie in $K'(t)[\Lambda, U]$; scalar extension cannot lower a reduced t-degree and thereby hide a larger degree over K'. Thus no K'-rational global separator, and in particular no $|K'| \gg C^2$ assumption, is used.

Lemma 8.8 (Nonseparating eliminant)**.** For every $\lambda \in K'^a$, including a colliding one,

$$q_\lambda(t, U) := \prod_{s=1}^{C} (U - \lambda(x_s(t))) = \chi_\Lambda(t, U)\big|_{\Lambda=\lambda}.$$

Every coefficient of q_λ has a common-denominator representation with numerator and denominator t-degrees at most D.

Proof. The first identity is the characteristic-polynomial product formula in the reduced generic fiber, with repeated factors allowed after specialization. The second assertion follows by substituting $\Lambda = \lambda$ in Ξ_Λ/Θ. Since Θ is independent of Λ, a separator collision cannot create a new parameter denominator or increase its degree. □

The coordinate interpolation numerators are already encoded in the same generic characteristic polynomial.

Lemma 8.9 (Characteristic-polynomial polar)**.** Define

$$v_j(t, U) = \sum_s x_{s,j}(t) \prod_{r \ne s} (U - \lambda(x_r(t))).$$

Then

$$v_j(t, U) = -\left.\frac{\partial \chi_\Lambda(t, U)}{\partial \Lambda_j}\right|_{\Lambda=\lambda}. \tag{74}$$

Consequently each v_j, and every fixed affine-linear combination of the v_j and q_U, has numerator and denominator t-degrees at most D, without assuming that λ separates the branches.

Proof. Over a splitting field, differentiate

$$\chi_\Lambda(t, U) = \prod_s \left(U - \sum_i \Lambda_i x_{s,i}(t) \right)$$

with respect to Λ_j and then specialize $\Lambda = \lambda$. This gives (74). Differentiating Ξ_Λ in the linear-form coefficients does not increase its t-degree, while the denominator Θ is unchanged. □

This directly covers the sole linear sketch used later:

$$b_1(P) = \lambda^{q^{d-1}} \cdot P.$$

Its interpolation numerator is the corresponding fixed K'-linear combination of the v_j.

For the rational oil filter

$$\psi(X) = \frac{H_0(X)}{c + \lambda(X)},$$

the preceding split-before-lift note proves that the saturated graph curve has degree at most $2D$. Apply the same construction to its generic linear element

$$u_{\Lambda,\mu} = \Lambda \cdot X + \mu Y,$$

where $Y = \psi(X)$ is the graph coordinate. Specializing $(\Lambda, \mu) = (\lambda, 0)$ gives q_λ, and

$$-\left.\frac{\partial \chi_{\Lambda,\mu}}{\partial \mu}\right|_{(\lambda,0)} = v_\psi.$$

The graph degree bound therefore gives numerator and denominator t-degrees at most $2D$ for the filter numerator. Consequently the existing precision

$$P = 4C_0 + 1$$

remains valid.

Remark 8.10. Separation is needed only when one wants the linear element to be a primitive coordinate for every point. The attack instead keeps the full characteristic polynomial and later accepts only a simple target factor. Characteristic polynomials and their coefficient polars remain defined at colliding linear forms.

8.5 Local separation at the target point

At target specialization, normalize q and each scalar numerator by the common t-valuation exactly as in Safey El Din–Schost Lemma 12 [27]. Call λ *valuation-generic* if every branch escaping to infinity with minimum coordinate valuation $\mu_s < 0$ satisfies

$$\operatorname{ord} \lambda(x_s) = \mu_s.$$

This condition, not pairwise branch separation, is what the specialization proof uses to remove the infinite factors from the specialized eliminant.

Lemma 8.11 (Simple-factor recovery)**.** Assume λ is valuation-generic. Let $\bar{q}(U)$ and $\bar{v}_b(U)$ be the normalized target specializations. If $\tau \in K'$ is a simple root of $\bar{q}$, then there is exactly one finite represented target point P with $\lambda(P) = \tau$, and

$$b(P) = \frac{\bar{v}_b(\tau)}{\bar{q}'(\tau)}.$$

No separation condition is needed away from that one fiber value.

Proof. Let S_∞ and S_0 index the escaping and finite branches. With

$$e = -\sum_{s \in S_\infty} \mu_s,$$

valuation genericity gives

$$(t^e q)(0,U) = \gamma \prod_{s\in S_0} (U - \lambda(P_s))$$

for a nonzero scalar γ. In the normalized numerator, the summand indexed by a finite branch s specializes to

$$\gamma b(P_s) \prod_{r\in S_0,\ r\neq s} (U - \lambda(P_r)).$$

A summand indexed by an escaping branch need not vanish term by term; rather, every such surviving summand is divisible by the full finite product $\prod_{r\in S_0}(U - \lambda(P_r))$. It therefore vanishes modulo $\bar{q}$, and in particular at every finite root of $\bar{q}$. Thus $\bar{v}_b$ modulo $\bar{q}$ is the ordinary interpolation numerator on the finite limits. A simple root τ belongs to exactly one finite branch, and evaluation followed by division by $\bar{q}'(\tau)$ gives the claimed value. □

Condition on the existence of a canonical eligible nonsingular base-field target point $P_\star$; for example, choose the first under any fixed ordering. Nonsingularity makes it an isolated multiplicity-one point. Let I generic branches escape to infinity and let F have finite target limits, counted with branch multiplicity. Then $I + F = C$. There are at most I valuation-cancellation hyperplanes and at most $F - 1$ collision hyperplanes against $P_\star$. Each bad equality

$$\lambda(w) = 0 \quad \text{or} \quad \lambda(P - P_\star) = 0$$

cuts out at most $|K'|^{a-1}$ choices of λ in $(K')^a$. Thus a uniform $\lambda \in (K')^a$ fails valuation genericity or collides at $P_\star$ with probability at most

$$\frac{I + F - 1}{|K'|} = \frac{C - 1}{|K'|}. \tag{75}$$

The coefficient vector of the equation may lie only over an algebraic closure, so its solution set need not literally be a K'-defined hyperplane. The count is still valid: choose a nonzero coordinate of that vector and fix all other coefficients of λ; the remaining equation has at most one solution in K' (and may have none).

For the filter denominator, first compute the C known start values $\lambda(\xi_s)$ and sample c uniformly from the complement of the forbidden set $\{-\lambda(\xi_s)\}_s$ in K'. Rejection sampling costs an expected factor at most $(1 - C/|K'|)^{-1}$ for a scan of the start values; this scan is negligible beside one length-P filter lift and is included in the miscellaneous envelope. Conditional on this choice, only the at most $F \le C$ finite target values remain forbidden. Therefore

$$\Pr[c + \lambda(P) = 0 \text{ for a finite target branch}] \le \frac{C}{|K'| - C}. \tag{76}$$

The old $C^2/|K'|$ event has disappeared.

8.6 Exact one-sketch Frobenius descent

Set

$$b_1(P) = \lambda^{q^{d-1}} \cdot P, \qquad K' = \mathbb{F}_{q^d}.$$

The separator value $\tau = \lambda(P)$ is already available from the factor of $\bar{q}$, so no numerator for $\lambda \cdot P$ is needed.

Lemma 8.12 (Simple-root descent is exact)**.** Let $\tau \in K'$ be a simple linear factor of the target eliminant, and let P be its unique represented point. Then

$$b_1(P)^q = \tau \iff P \in \mathbb{F}_q^a.$$

Proof. The forward identity for a base-field point is immediate. Conversely,

$$b_1(P)^q = \lambda \cdot P^q.$$

If it equals $\tau = \lambda \cdot P$, then P and P^q have the same separator value. The target equations and homotopy are defined over $\mathbb{F}_q$, so the represented finite target limits are stable under Frobenius. If $P^q \neq P$, the two distinct represented points produce multiplicity at least two at $U = \tau$, contradicting simplicity. Hence $P^q = P$. $\square$

This eliminates the probabilistic $C/|K'|$ false-positive term. It also eliminates the abort cap: every simple factor passing the Frobenius test and the nonzero-square oil filter is a genuine base-field target solution. The algorithm reconstructs coordinates for the first such factor and stops, so exactly one coordinate re-lift is charged per successful forgery.

8.7 Root success probability

For completeness, this pass expands the probability calculation inherited from the split-before-lift note. Let M count all off-oil affine roots of the normalized residual system, and let N count those with invertible affine Jacobian. The parent manuscript proves, for an actual parameter $\mu \geq 1 - q^{-1}$,

$$\mathbb{E}[M] = \mu, \qquad \mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}, \qquad \mathbb{E}[N] = \mu\pi,$$

where $\pi = \prod_{j=1}^{r-1}(1 - q^{-j})$. The roots occur in pairs $z, -z$, and each nonsingular normalized pair yields one eligible projective point. Applying the second Bonferroni inequality to the projective-point events and bounding pairs of nonsingular roots by pairs of all roots gives

$$\begin{aligned} \Pr[\text{eligible nonsingular projective root}] &\geq \frac{\mathbb{E}[N]}{2} - \mathbb{E}\left[\binom{M/2}{2}\right] \\ &= \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8}. \end{aligned}$$

The right-hand side is increasing for $\mu \in [1 - q^{-1}, 1]$, so substituting $\mu = 1 - q^{-1}$ yields

$$p_{\text{root}} = 0.3690869822\ldots.$$

As a separate fallback, the ledger also substitutes the parent manuscript's already-proved lower bound $0.326336998767509\ldots$; the Level-I total then becomes 140.531461, still 2.469 bits below target.

8.8 Finite Level-I ledger

At Level I the local variant has two scalar families, b_1 and ψ, instead of three. The terminal product tree therefore uses

$$\frac{a}{2}(1 + 2 \cdot 2) = 125$$

long-product units. Rational reconstruction contributes 99.885 units. Candidate coordinate evaluation is charged once over the complete Las Vegas execution, rather than per trial, and the 64-unit miscellaneous cushion is retained. This pass also charges the one-time formation of $u_s = \lambda \cdot x_s$ during that coordinate rerun.

The one-trial success lower bound is 0.353997, giving restart factor 2.824886. At $d = 8$, the local-separator failure bound in (75) is 0.016637, and the conditional target-denominator bound in (76) is 0.016918.

Table 2: Level-I component ledger for extension degree $d = 8$.

Component	$\log_2$ gates
Initial base-field branch lifts	139.358
Valuation-safe filter series	132.120
Terminal scalar recombination	138.791
One on-demand coordinate re-lift	137.860
Extension-valued leaf formation	127.610
Coordinate-rerun separator leaves	124.898
Factorization and exact verification envelope	115.707
Preprocessing	63.365
Total	**140.383995**

Table 3: Level-I sensitivity to the multiplier on $\mathsf{M}(n)\log n$ in rational reconstruction, retaining $d = 8$.

Multiplier	8	10	16	32	64	128
Total	140.384	140.424	140.538	140.805	141.225	141.815

Table 4: Level-I cross-checks. Each row reoptimizes the extension degree.

Online-envelope multiplier	1	2	4	8	16
Best extension degree	8	8	9	9	9
Total	140.384	141.117	141.952	142.856	143.806

Thus the candidate remains below 2^{143} throughout the rational-reconstruction sensitivity range. It also remains below target after multiplying the entire claimed online-product envelope by eight, but not by sixteen; the finite margin therefore does not excuse the remaining obligation to provide a fully explicit schedule. The aggregate Level-I margin corresponds to an overall multiplicative cushion of $2^{2.616} = 6.13$ in the inherited gate model; this does not address memory traffic or a change of model.

9 Evidence and limitations

The direct and directional reductions, the fixed-output scissors theorem, the determinant identities used by the local separators, and the exact Frobenius filter are algebraic results. They do not depend on the final Boolean-gate convention. Every reconstructed candidate is checked against the original public equations or the unchanged public verifier. A failed solver invocation may therefore force a retry, but it cannot produce an accepted false signature or equivalent key.

The reduced implementation shows that the local-separator attack composes at toy scale. It preserves the QR-UOV public quadratic relation but simplifies seed expansion, byte serialization, and the exact domain-separated hash interface. Starting from expanded public matrices, a message, and a salt, it runs the complete attack and produces a signature accepted by the public verifier. A public-only replay reproduces the result. The canonical run and five separate regressions give six successful $q = 13$ forgeries and replays. One supplementary $q = 29$ instance also forged and replayed successfully. This closes the composability gap between the individual algebraic checks, but it does not establish the full-parameter Boolean-gate estimate. The prototype uses naive truncated-series arithmetic at $a = 2$, so it neither implements nor benchmarks the fast multiplication schedule used in that estimate. The full experiment record appears in Section F.

That estimate remains conditional on four points. First, the characteristic-polynomial degree theorem must apply to the exact saturated dominant homotopy algebra and its rational-filter graph with the stated degree bounds. Second, the fast online-product envelope needs a line-by-line full-scale schedule. Third, the public-random model must transfer to the seeded public-key expansion. Finally, the gate ledger does not account for the astronomical memory traffic of the symbolic representation. No defect is currently known in the determinant identities, but the Level-I margin is only 2.616 bits, so these obligations are material. The technical supplement retains the complete validation record and executable-audit inventory.

Table 5: Status of the main claims.

Claim	Status
Projective direct core $m-4$	Exact for every nonzero target once a seed root is supplied; Chevalley–Warning guarantees such a root in every seven-dimensional seed, and homogeneity quotients the residual by scaling.
Seed root is off oil	Fixed-key probability bound over public attack coins; no random-key assumption.
Common-isotropic extension	Deterministic and finite after the seed, using linear algebra and exhaustive search in $\mathbb{P}^6(\mathbb{F}_q)$.
Eligible projective-root probability	Exact lower bound in the independent-graph public-random degree-three model.
One regular direction determines an equivalent key	Exact for every expanded key satisfying the public rank and signing checks.
Returned signature or key is valid	Exact; every candidate is verified against the unchanged public map.
Finite cost rows	Reproducible symbolic-algorithm diagnostics, not measured circuits.
Full-parameter memory	Prohibitive for the direct representations analyzed here.
Fixed SHAKE/AES expansion	Key-by-key verification is exact; density statements transfer only in the stated ideal-primitive models, with explicit buffer and seed-collision terms.

10 Combined cost, memory, and parameter implications

10.1 Work unit and scope

The QR-UOV specification expresses classical attack costs as Boolean-gate exponents and charges multiplication of b-bit field elements by

$$\mathsf{G}_{\mathrm{mul}}(b) = 2b^2 + b \tag{77}$$

Boolean gates [2]. We retain that broad unit. Every extension-field operation is overcharged as one multiplication using the explicit Karatsuba circuit of (72).

The principal symbolic-homotopy term uses the corrected reconstruction precision $2C_0$. For a square solver core of dimension a,

$$C = 2^a, \qquad C_0 = 2^{a-1}(a+2),$$

and the flattened lifting schedule is charged by

$$3(a^3 + a^2)\mathsf{M}(8CC_0) \tag{78}$$

operations in the chosen working field. The projective direct row uses $a = m-4$ and includes the fixed-key seed bound, nonsingular-root factor, random chart, and finite-field separator. The

Table 6: Classical gate diagnostics, including the local-separator split-before-lift refinement and the audited Karatsuba field circuit. A positive margin means that the attack row lies below the category target.

	Level I		Level III		Level V	
Attack	Cost	Margin	Cost	Margin	Cost	Margin
Original projective forgery	151.639	−8.639	203.507	3.493	260.389	11.611
Local-separator projective forgery	140.384	2.616	191.027	15.973	247.125	24.875
Directional equivalent-key recovery	154.826	−11.826	206.177	0.823	260.857	11.143
Best attack	**140.384**	**2.616**	**191.027**	**15.973**	**247.125**	**24.875**
Category target	143		207		272	

directional row uses $a = V$ and includes direction regularity and its separator. Polynomial-time preprocessing, all-equation filtering, factorization, graph completion, square testing and rescaling, and public verification are included through explicit lower-stage envelopes; they do not change the displayed decimals.

These are reproducible diagnostics for symbolic algorithms, not measured circuits. In particular, setting

$$\mathsf{M}(t) = t \log_2 t \log_2 \log_2 t$$

assigns unit coefficient to an asymptotic multiplication bound. The exact reductions, Karatsuba circuits, sparse fields, and success probabilities do not depend on that convention; the final conversion to a practical implementation does.

10.2 The two attacks

The original direct rows use projective solver dimensions $50, 74, 101$. The local-separator refinement keeps those algebraic cores but lowers the working-field degrees and removes global branch separation, the second Frobenius numerator, probabilistic false candidates, and repeated coordinate recovery. Its optimized field degrees are 8, 12, and 16 in the supplied ledger. The directional rows remain an independent equivalent-key route. The reduced implementation executes the local-separator route at toy scale; the full-parameter symbolic lift still dominates both time and memory.

For context, the QR-UOV team's January 2026 response reports classical Hashimoto estimates of $158, 217, 285$ bits for the same three public dimensions [29]. The new direct rows are lower by approximately $6.36, 13.49, 24.61$ bits. This is not a claim that the implementations are identical: the team uses a semi-regular direct-MQ model, whereas our row uses the finite-characteristic nonsingular-root solver and its explicit success probability.

10.3 What follows at each level

Level I. The local-separator row is 140.384, or 2.616 bits below the 143-bit target. Thus the paper gives a candidate Level-I break in the stated Boolean-gate model. This is not a practical attack claim: the symbolic state is physically unrealizable, and the full-scale online-product schedule and characteristic-polynomial degree transfer remain explicit obligations. The reduced-parameter implementation establishes that the algebraic pipeline is a single executable attack, not that the Level-I extrapolation is practical.

Level III. The local-separator row is 191.027, leaving 15.973 bits below target. The older projective and independent equivalent-key rows are also below target. The combined evidence

Table 7: Smallest quotient-compatible (V, m) pairs found by the companion audit that put both displayed attack rows at the category target, or at the target plus a 16-bit attack-specific cushion.

	Current		Reach target		Reach target +16	
Level	V	m	V	m	V	m
I	52	54	52	54	57	60
III	76	78	78	81	87	90
V	102	105	108	111	117	120

strongly supports changing the Level-III parameters.

Level V. The local-separator row is 247.125, leaving 24.875 bits below target. The older projective and independent equivalent-key rows are likewise below target. Level V should not be retained unchanged.

10.4 Why changing only the vinegar count is not a repair

The official parameter sets sit on the direct side of the projective scissors crossing:

Level	V	$m-4$	Directional core	Projective direct core
I	52	50	52	50
III	76	74	76	74
V	102	101	102	101

Increasing V raises the directional core but leaves the projective direct core unchanged. Decreasing V eventually makes directional recovery no larger than $m-4$. Consequently a fixed-m vinegar retuning cannot move the combined algebraic bottleneck.

The next table gives attack-specific thresholds under the same padded ledger. It is not a complete replacement proposal: every other attack, failure probability, key size, and implementation cost must be recomputed. The point is that both columns must move. Raising only V leaves the direct attack unchanged; raising only m leaves the directional attack unchanged.

The table enforces $m \equiv 0 \pmod 3$ and the direct-side feasibility condition $V \geq m-3$. Level I already exceeds its bare target in this model, but a 16-bit cushion requires moving both dimensions. At Levels III and V, even the target-only diagnostic requires increases in both V and m; a 16-bit cushion requires substantially larger changes. A structural redesign may be preferable to paying both costs.

10.5 Memory remains the central practical limitation

The desired outputs are small: one direct attack needs a single vector in $\mathbb{F}_q^n$, and the equivalent-key attack ultimately needs a $V \times O$ graph. The direct symbolic representations are not small. One flattened coefficient array at terminal precision requires approximately

$$2^{49.41}\ \text{EiB}, \qquad 2^{98.51}\ \text{EiB}, \qquad 2^{153.40}\ \text{EiB}$$

for the projective direct cores. The corresponding directional arrays require about $2^{53.73}$, $2^{102.68}$, and $2^{155.41}$ EiB. A full-parameter execution using these representations is physically unrealistic.

This limitation prevents a claim of a practical full-parameter implementation. No full-parameter forgery or equivalent-key recovery is reported, and the paper draws no conclusion from unimplemented attack variants. The exact reductions and the fixed-output scissors theorem are algebraic results; time, storage, and correctness are reported as separate resources.

11 Conclusion

QR-UOV's projective direct-forgery and directional equivalent-key attacks obey a fixed-output algebraic ceiling. For every vinegar count and every nonzero verification target, one of the two attacks has nonlinear solver core dimension at most $m-4$. Changing only the vinegar count therefore cannot eliminate both low-dimensional attack surfaces. A revision must also increase the output dimension or change the construction so that one of the reductions no longer applies.

The direct reduction gives square cores of dimensions $50, 74, 101$, while the directional attack recovers an equivalent signing key. To estimate the cost of these cores, the local-separator refinement removes the need for global branch separation: characteristic-polynomial polars suffice for scalar recovery, and one exact Frobenius sketch identifies a base-field point on a simple target factor. Under the stated symbolic Boolean-gate model, the resulting rows lie below all three NIST category targets, including Level I by 2.616 bits.

The reduced-parameter implementation now composes the complete attack and produces publicly accepted forgeries from public inputs alone. Consequently, the result is no longer merely a collection of independently tested micro-lemmas. What remains conditional is the full-parameter complexity projection: the exact characteristic-polynomial degree transfer to the saturated dominant algebra, the fast online-product schedule, seeded-instance transfer, and memory traffic all require further scrutiny. The paper therefore claims a finite, executable structural attack and a model-based candidate full-parameter break, not a practical full-parameter computation.

The algebraic ceiling is exact, while the concrete bit-security estimates are conditional on the stated solver and cost model. The reduced-parameter implementation establishes that the attack components compose. Any QR-UOV revision should reconsider both dimensions and the underlying compressed structure rather than apply a vinegar-only retuning.

References

[1] H. Furue, Y. Ikematsu, Y. Kiyomura, and T. Takagi. A new variant of Unbalanced Oil and Vinegar using quotient ring: QR-UOV. Cryptology ePrint Archive, Paper 2020/1243, 2020.

[2] H. Furue, Y. Ikematsu, F. Hoshino, T. Takagi, H. Kosuge, K. Yamakoshi, R. Akiyama, S. Nakamura, S. Orihara, and K. Kinjo. *QR-UOV Specification Document (Round 2 submission).* NIST Additional Digital Signatures Project, 5 February 2025.

[3] A. Kipnis, J. Patarin, and L. Goubin. Unbalanced Oil and Vinegar signature schemes. In *EUROCRYPT 1999*, pp. 206–222, 1999.

[4] National Institute of Standards and Technology. Nine candidates advance to the third round of the Additional Digital Signatures for the PQC standardization process. NIST CSRC, 14 May 2026.

[5] G. Alagic et al. Status report on the second round of the Additional Digital Signature Schemes for the NIST Post-Quantum Cryptography Standardization Process. NIST IR 8610, May 2026.

[6] O. Benoist. Degrés d'homogénéité de l'ensemble des intersections complètes singulières. *Annales de l'Institut Fourier*, 62(3):1189–1214, 2012.

[7] J. van der Hoeven and G. Lecerf. On the complexity exponent of polynomial system solving. *Foundations of Computational Mathematics*, 21(1):1–57, 2021.

[8] M. Giusti, G. Lecerf, and B. Salvy. A Gröbner free alternative for polynomial system solving. *Journal of Complexity*, 17(1):154–211, 2001. doi:10.1006/jcom.2000.0571.

[9] N. Giménez, J. Heintz, G. Matera, L. M. Pardo, M. Pérez, and M. Privitelli. A Kronecker algorithm for locally closed sets over a perfect field. arXiv:2512.14888v2, 5 June 2026.

[10] W. Beullens. Improved cryptanalysis of UOV and Rainbow. In *EUROCRYPT 2021, Part I*, pp. 348–373, 2021.

[11] H. Furue and Y. Ikematsu. A new security analysis against MAYO and QR-UOV using rectangular MinRank attack. In *IWSEC 2023*, pp. 101–116, 2023.

[12] T. Suzuki, H. Furue, T. Ito, S. Nakamura, and S. Uchiyama. An extended rectangular MinRank attack against UOV and its variants. In *IWSEC 2025*, LNCS 16208, pp. 111–130, Springer, 2026.

[13] H. Furue and Y. Ikematsu. Key recovery attacks on UOV using p^ℓ-truncated polynomial rings. Cryptology ePrint Archive, Paper 2026/298, 2026.

[14] A. May, M. Ostuzzi, and H. Ressler. Just Guess: Improved (quantum) algorithm for the underdetermined MQ problem. In *EUROCRYPT 2026, Part IV*, LNCS 16544, pp. 334–362, 2026.

[15] P. Pébereau. One vector to rule them all: Key recovery from one vector in UOV schemes. In *PQCrypto 2024*, pp. 92–108, 2024; ePrint 2023/1131.

[16] W. Baur and V. Strassen. The complexity of partial derivatives. *Theoretical Computer Science*, 22(3):317–330, 1983. doi:10.1016/0304-3975(83)90110-X.

[17] C. D'Andrea and A. Dickenstein. Explicit formulas for the multivariate resultant. *Journal of Pure and Applied Algebra*, 164(1–2):59–86, 2001. doi:10.1016/S0022-4049(00)00145-6.

[18] G. Jeronimo, T. Krick, J. Sabia, and M. Sombra. The computational complexity of the Chow form. *Foundations of Computational Mathematics*, 4(1):41–117, 2004. doi:10.1007/s10208-002-0078-2.

[19] V. Neiger, B. Salvy, E. Schost, and G. Villard. Faster modular composition using two relation matrices. In *ISSAC 2026*, pp. 315–324, 2026; arXiv:2601.17422.

[20] S. Jaques, L. Ran, S. Samardjiska, and M. Seitner. Smoothing the degree of regularity for polynomial systems. Cryptology ePrint Archive, Paper 2026/408, 2026.

[21] R. Kouba, V. Neiger, and M. Safey El Din. A complexity analysis of the F4 Gröbner basis algorithm with tracer data. arXiv:2603.16378, 2026.

[22] E. Kaltofen and V. Shoup. Subquadratic-time factoring of polynomials over finite fields. *Mathematics of Computation*, 67(223):1179–1197, 1998.

[23] D. Wiedemann. Solving sparse linear equations over finite fields. *IEEE Transactions on Information Theory*, 32(1):54–62, 1986.

[24] J. Heintz. Definability and fast quantifier elimination in algebraically closed fields. *Theoretical Computer Science*, 24:239–277, 1983.

[25] G. Lachaud and R. Rolland. On the number of points of algebraic sets over finite fields. *Journal of Pure and Applied Algebra*, 219(11):5117–5136, 2015.

[26] M. O. Rabin. Probabilistic algorithms in finite fields. *SIAM Journal on Computing*, 9(2):273–280, 1980. doi:10.1137/0209024.

[27] M. Safey El Din and E. Schost. Bit complexity for multi-homogeneous polynomial system solving: Application to polynomial minimization. arXiv:1605.07433v2, 2017; manuscript revised 6 November 2018.

[28] E. Schost. Computing parametric geometric resolutions. *Applicable Algebra in Engineering, Communication and Computing*, 13(5), 2002. doi:10.1007/s00200-002-0109-x.

[29] S. Orihara, on behalf of the QR-UOV team. Round 2 (Additional Signatures) official comment: QR-UOV. NIST PQC Forum, 20 January 2026.

[30] Y. Hashimoto. An improvement of algorithms to solve under-defined systems of multivariate quadratic equations. *JSIAM Letters*, 15:53–56, 2023. doi:10.14495/jsiaml.15.53.

[31] E. Thomae and C. A. Wolf. Solving underdetermined systems of multivariate quadratic equations revisited. In *PKC 2012*, LNCS 7293, pp. 156–171, 2012. doi:10.1007/978-3-642-30057-8_10.

[32] H. Furue, S. Nakamura, and T. Takagi. Improving Thomae–Wolf algorithm for solving underdetermined multivariate quadratic polynomial problem. In *PQCrypto 2021*, LNCS 12841, pp. 65–78, 2021. doi:10.1007/978-3-030-81293-5_4.

[33] C. Chevalley. Démonstration d'une hypothèse de M. Artin. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 11:73–75, 1935.

[34] E. Warning. Bemerkung zur vorstehenden Arbeit von Herrn Chevalley. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 11:76–83, 1935.

[35] Y. Jin, Y. Pan, X. He, B. Gong, and J. Ding. Security analysis on UOV families with odd characteristics: Using symmetric algebra. Cryptology ePrint Archive, Paper 2025/1137; in *PKC 2026, Part I*, LNCS 16551, pp. 428–454, 2026.